NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR306527
Lead Applicant	Dr Mayur Murali
Research Title	BREATHE: Better Respiratory wEAning using Transparent macHine learning and data visualisation for invasivE mechanical ventilation
Plain English Summary	**Aim** Improve the care of patients in intensive care who need help with their breathing. **Background** Every year, 70,000 patients in intensive care need help with breathing. This is delivered through a machine called a ventilator. To use a ventilator properly, clinicians must understand the patient and their disease. A clinical decision support system (CDSS) is a type of computer software that helps clinicians make better decisions. It matches patient and disease features to a database. Treatment options are then suggested to clinicians. The NHS Long Term Plan has suggested CDSS should be used to help care for patients. Deciding when to reduce ventilator support is important. Leaving patients on a ventilator too long can cause lung injury. However, patients must be strong enough to breathe on their own. Previous research has shown differences in clinical practice when reducing ventilator support. There is mixed success and no agreement on best practice. This suggests CDSS could improve ventilator care. Our national survey showed CDSS are not used for ventilator management. This is because there is limited ready-to-use computer software. The survey also showed clinicians do not trust CDSS. This is because their decisions are rarely explained. **Addressing the problem** This research aims to address these two problems. 1. Use computer-based methods to find patterns in patient data. This means studying lots of information from many patients. This method is called machine learning (ML). 2. Create new ways to show the data using visual tools. This will help people understand, trust and use CDSS better. **Design and methods** This research will be done in three steps. 1. Understanding the patient. We will use ML to predict when to reduce ventilator support. Predictions will cover the patient's whole stay in the hospital. They will show the key reasons for each prediction. We will check how well the models work and make sure they are fair for all groups. This includes those less often studied. 2. Designing new ways to show predictions using visual tools. Clinicians who work in intensive care will help create these tools, so they are helpful. The tools will show predictions and explain the important reasons behind them. This will be shown on an interactive computer system. 3. Checking the impact on care. We will test whether these tools make decisions harder or easier by comparing them to normal practice. We will do this through simulation testing. Our findings will guide future work. This could include testing CDSS in intensive care. **Patient and public involvement** This project has involved patients and the public from the start. This has shown this research can make a big difference in four ways: 1. Make clinical care more consistent. 2. Improve patient safety. 3. Help clinicians share information with patients and families using visual tools. 4. Help the NHS by shortening hospital stays and lowering costs. This work focuses on two important areas from the James Lind Alliance. These have been chosen through discussion with patients. **Sharing our findings**

	I will share our findings through local scientific meetings and conferences. Results will be published in academic journals. We will use Imperial College London and collaborator networks to promote our work. This will be seen by industry, other research groups, and stakeholders. I have the support of a patient panel to help plan and share our work with the public. This will be supported by the Imperial College Science Media Team and public activities and events.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

The NIHR Academy is part of the Department for Health and Social Care (DHSC), National Institute for Health and Care Research (NIHR). The contracting agent for NIHR Academy is the Leeds Teaching Hospital Trust (LTHT). The DHSC is the Data Controller and LTHT is the Data Processor under the General Data Protection Regulation (GDPR) EC 2016/679. DHSC NIHR respects the privacy of individuals who share their data and processes it in a manner that meets the requirements of GDPR. The DHSC Data Protection Officer can be contacted by email at: data_protection@dhsc.gov.uk

The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

Table Of Contents

1. Application Summary Information
2. Applicant CV
3. Applicant Research Background
4. Plain English Summary of Research
5. Scientific Abstract
6. Detailed Research Plan
7. Patient & Public Involvement
8. Training & Development and Research Support
9. Detailed Budget
10. Uploads
11. Participants and Signatory
12. Acknowledgement, Review and Submit

1. Application Summary Information

Host organisation (which will administer any award)
Imperial College of Science, Technology and Medicine

Partner organisation (if applicable)	

Application Type

Strategic Theme	Health data science

Partnership Fellowship	N/A

WTE % Please select which WTE option you would prefer
1.00

Duration Doctoral Fellowship is a 3 year award and based on your WTE preference the duration in months is calculated as follows
36

Research Title	BREATHE: Better Respiratory wEAning using Transparent macHine learning and data visualisation for invasivE mechanical ventilation

Research Type	Pilot study, Secondary data analysis, Qualitative study

Proposed start date if grant awarded	01/10/2025

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

The 'Publication Record' section of the form is automatically populated from publications added into the "My Research Outputs" page of your ARAMIS account.

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
09/2022	11/2024	MSc	Health Data Science	United Kingdom	University College London	Pass with Merit	Institute of Health Informatics
02/2017	12/2021	FRCA	Fellowship of the Royal College of Anaesthetists	United Kingdom	The Royal College of Anaesthetists	Pass	The Royal College of Anaesthetists
02/2016	05/2016	DTM&H	Diploma in Tropical Medicine & Hygiene	United Kingdom	Liverpool School of Tropical Medicine	Pass with Merit	School of Tropical Medicine
10/2015	07/2019	MRCP(UK)	Membership of the Royal College of Physicians	United Kingdom	Royal College of Physicians	Pass	Royal College of Physicians
09/2014	08/2018	MSc	Masters in Medical Education	United Kingdom	University College London	Pass with Merit	UCL Medical School
09/2009	07/2013	MBBS(Hons)	Medicine	United Kingdom	Queen Mary University of London	Pass with Merit in Clinical Specialties	Faculty of Medicine & Dentistry
09/2005	07/2008	BSc(Hons)	Biological Sciences with Virology	United Kingdom	University of Warwick	2(i) Hons (Upper Second Class)	School of Life Sciences

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
11/2024	-	ST7 - Anaesthetics	Northwick Park Hospital	0
08/2024	11/2024	ST7 - Intensive Care Medicine	Imperial College Healthcare NHS Trust - St Mary's Hospital	0
05/2024	08/2024	ST6 - Anaesthetics/ Intensive Care Medicine	National Hospital for Neurology & Neurosurgery	0
02/2024	05/2024	ST6 - Anaesthetics/ Intensive Care Medicine	Royal Brompton Hospital	0
08/2023	02/2024	ST6 - Anaesthetics	Chelsea & Westminster Hospital	0
08/2022	08/2023	ST5 - Intensive Care Medicine	Barts Health - Royal London Hospital	0
05/2022	08/2022	ST4 - Academic Clinical Fellow	Imperial College Healthcare NHS Trust	100
02/2022	05/2022	ST4 - Intensive Care	Barts Health - Royal London	0

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
		Medicine	Hospital	
08/2021	02/2022	ST4 - Academic Clinical Fellow	Imperial College Healthcare NHS Trust	100
05/2021	08/2021	ST3 - Anaesthetics	Imperial College Healthcare NHS Trust - Hammersmith Hospital	0
08/2020	05/2021	ST3 - Anaesthetics	Imperial College Healthcare NHS Trust - St Mary's Hospital	0
02/2020	08/2020	Trust Grade Registrar	Imperial College Healthcare NHS Trust - St Mary's Hospital	0
08/2019	02/2020	Junior Fellow	Zambia Anaesthesia Development Program	0
08/2018	08/2019	National Medical Director's Clinical Fellow	NHS England	0
08/2017	08/2018	CT2 - Anaesthetics	Barts Health - Whipss Cross University Hospital	0
08/2016	08/2017	CT1 - Anaesthetics	University College London Hospital	0
05/2016	08/2016	Trust Grade SHO	Barts Health - Whipps Cross University Hospital	0
08/2015	02/2016	Junior Clinical Fellow	St George's University Hospital	0
08/2014	08/2015	F2	Whipps Cross University Hospital	0
08/2013	08/2014	F1	Southend University Hospital	0

Research Grants held

Publication Record
(include the list of authors, article title, journal name, and volume)
This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Murali, M. and Agarwal, S.. (2024) ‘Wellbeing: paved with good intentions, but the road needs fixing first.’, *Anaesthesia*.
Thomas, MAF., Ward, CJ., Sinnott, ME., Davies, TW., Wong, JM., Wong, JKL., Kunst, G. and Anwar, S.. (2024) ‘Nitrous Oxide Manifold and Other Reduction of Emissions (NoMoreGas): a multicentre observational study evaluating pre-utilisation loss of nitrous oxide.’, *British journal of anaesthesia*.
Campbell, RAS., Thevathasan, T., Wong, DJN., Wilson, AM., Lindsay, HA., Campbell, D., Popham, S., Barneto, LM., Myles, PS., Moonesinghe, SR. and Harris, SK.. (2024) ‘Critical care unit bed availability and postoperative outcomes: a multinational cohort study.’, *Anaesthesia*, 79(11).
Murali, M. and Wiles, MD.. (2024) ‘Large language models and artificial intelligence: the coming storm for academia.’, *Anaesthesia*.
Sunkersing, D., Goodfellow, H., Mu, Y., Ramasawmy, M., Murali, M., Adams, L., FitzGerald, TJ., Blandford, A., Stevenson, F., Bindman, J., Robson, C. and Banerjee, A.. (2024) ‘Long COVID symptoms and demographic

Publication Record

(include the list of authors, article title, journal name, and volume)
This information can be edited via the "My Research Outputs" page of your ARAMIS account.

Publications
associations: A retrospective case series study using healthcare application data.', *JRSM open*, 15(7).
Murali, M., Ni, M., Karbing, DS., Rees, SE., Komorowski, M., Marshall, D., Ramnarayan, P. and Patel, BV.. (2024) 'Clinical practice, decision-making, and use of clinical decision support systems in invasive mechanical ventilation: a narrative review.', *British journal of anaesthesia*, 133(1).
Murali, M. and Charlesworth, M.. (2024) 'How to write pilot and feasibility studies.', *Anaesthesia*, 79(6).
Murali, M. and Inada-Kim, M.. (2023) 'Early warning scores: the case for aggregate vs. single extreme parameter activation to detect patient deterioration.', *Anaesthesia*, 78(7).
Murali, M., Gumber, L., Jethwa, H., Ganesh, D., Hartmann-Boyce, J., Sood, H., Zaccardi, F. and Khunti, K.. (2023) 'Ethnic minority representation in UK COVID-19 trials: systematic review and meta-analysis.', *BMC Medicine*, 21(1).
Ramasawmy, M., Poole, L., Thorlu-Bangura, Z., Chauhan, A., Murali, M., Jagpal, P., Bijral, M., Prashar, J., G-Medhin, A., Murray, E., Stevenson, F., Blandford, A., Potts, HWW., Khunti, K., Hanif, W., Gill, P., Sajid, M., Patel, K., Sood, H., Bhala, N., Modha, S., Mistry, M., Patel, V., Ali, SN., Ala, A. and Banerjee, A.. (2022) 'Frameworks for Implementation, Uptake, and Use of Cardiometabolic Disease-Related Digital Health Interventions in Ethnic Minority Populations: Scoping Review.', *JMIR cardio*, 6(2).
(2021) 'Davies RL, Murali M. Medical leadership in the military and NHS: what transferable lessons can be identified?BMJ Leader 2021; 5: 232-235.',
Odor, PM., Bampoe, S., Lucas, DN., Moonesinghe, SR., Andrade, J. and Pandit, JJ.. (2020) 'Protocol for direct reporting of awareness in maternity patients (DREAMY): a prospective, multicentre cohort study of accidental awareness during general anaesthesia.', *International journal of obstetric anesthesia*, 42.
Mole, G., Murali, M., Carter, S., Gore, D., Broadhurst, J., Moore, T., Vickers, P. and Miles, A.. (2019) 'A service evaluation of specialist nurse telephone follow-up of bowel cancer patients after surgery.', *British journal of nursing (Mark Allen Publishing)*, 28(19).
Murali, M., Ly, C., Tirlapur, N., Montgomery, HE., Cooper, JA. and Wilson, AP.. (2019) 'Diarrhoea in critical care is rarely infective in origin, associated with increased length of stay and higher mortality.', *Journal of the Intensive Care Society*, 21(1).
Wong, DJN., Harris, SK. and Moonesinghe, SR.. (2018) 'Cancelled operations: a 7-day cohort study of planned adult inpatient surgery in 245 UK National Health Service hospitals.', *British journal of anaesthesia*, 121(4).
Murali, M.. (2017) 'Tackling global medical education inequality through e-learning.', *Medical education*, 51(5).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/09/2020	Imperial School of Anaesthesia	First Prize – Imperial School of Anaesthesia Trainee Presentation Day	Presented project on echo education in Zambia
01/07/2019	Royal Society of Medicine, Anaesthesia Section	Second Prize – Anaesthesia Section Trainee Prize	Presented project on critical incident management
01/07/2019	Association of Anaesthetists	IRC Travel Grant	Primary applicant
01/06/2019	Society for Education in Anaesthesia (UK)	Society for Education in Anaesthesia Travel Grant	Primary Applicant
01/06/2019	Beit Trust	Beit Bursaries for Junior Doctors (travel grant)	Primary Applicant

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/06/2018	Pan London Trauma Symposium	First Prize for Innovation	Led QI on silver trauma management
01/07/2014	Southend University Hospital	Judi Sharpe Scholarship for Teaching and Educational Excellence	F1
01/07/2014	North East Thames Foundation School	Teaching Excellence Award (Trainee Merit Awards in Foundation)	F1
01/06/2013	Newham Hospital	Teaching Excellence Award	Medical student
01/04/2013	Barts & The London School of Medicine & Dentistry	Enid Linder Foundation Elective Bursary	Medical student
01/10/2012	Medical Council on Alcohol	First Place – Michael Frowen Memorial Essay Prize for Medical Students	Medical student
01/06/2012	Barts & The London School of Medicine & Dentistry	Merit in Clinical Specialties	Medical student

ORCID iD 0000-0002-2340-7810

3. Applicant Research Background

Applicant Professional Background	Medically Qualified

Specialty (if Medically Qualified)
Anaesthetics

Do you have a GMC number?	Yes

GMC number Please supply a valid GMC number	7410240

Please describe your research career to date

My research career has involved developing skills in data science, health services and qualitative research. To date, I have 12 academic publications, 12 conference abstracts, and have co-edited and published a book. The next stage of my academic development is to train to become an expert in these disciplines in my area of academic interest, invasive mechanical ventilation (IMV) in critical care.

Research Interests

During my Academic Clinical Fellowship at Imperial College London, I developed research interests and learned from academic supervisors and collaborators. Background work has included a narrative review assessing ventilation practice and decision-making (published in the *British Journal of Anaesthesia*); a national survey assessing use of clinical decision support systems (CDSS) in intensive care; and preliminary work building a machine learning model using ventilation data from a randomised controlled trial of over 200 patients with collaborators at Aalborg University. I have developed skills in data science, completing a Masters in Health Data Science at UCL while working full-time clinically, for which I was awarded a Merit. I have learned to code in Python, SQL, Matlab, Stata and R, and gained experience of statistics and machine learning. I am a national coordinator of GEMINI, an international study evaluating ventilation practice, through which I have experienced the Health Research Authority approval process and collaborated with NHS Trusts to contribute to a database of over 10,000 patients, which will act as a data repository for my PhD. The first abstract from GEMINI will be presented at the American Thoracic Society Conference in 2025.

I have developed collaborations with the Imperial-NIHR HealthTech Research Centre, with support provided by methodologists, health economists and implementation scientists; the Imperial Data Science Institute, to assist with machine learning analysis and visualisation development; and Aalborg University, drawing on their expertise in CDSS and biomedical engineering. For GEMINI, I have collaborated with the VentilaGroup in Spain. Additionally, background work has involved collaboration with the Intensive Care Society and the Paediatric Critical Care Society.

I have collaborated with patients through presentations workshops organised by the Research Partner's Group and the Patient Experience Research Centre at Imperial. I have recruited a patient co-applicant and convened a Patient and Family Advisory Group to advise on the project through the charities ICUSteps and Voice. These experiences have refined the proposal, lay summary and outcome measures.

Education & Qualitative Research Experience

I have completed a Masters in Medical Education (UCL), achieving a Merit. Using these skills, I developed my own teaching, winning prizes and being awarded Fellowship of the Higher Education Academy. I developed the Health Professions Education Masters course with Kings College London for healthcare students in Somaliland, teaching virtually and in person; undertook a teaching fellowship in Zambia, for which I was awarded several competitive travel bursaries, delivering simulation and skills workshops, the impact of which have been presented internationally; and co-written online modules for BMJ Learning ("Challenges in Hospital Care: Sepsis –

recognition and management in adults") and the Association of Anaesthetists ("Professional Practice in Resource Poor Settings"). I co-edited and wrote several chapters for a textbook, "Surviving Prescribing", published by Cambridge University Press. I am on the committee of two courses delivered through the Imperial School of Anaesthesia, the Novice Course, and the Imperial Final Viva Course. Through the Masters, I developed critical appraisal and qualitative analysis skills, and learned how to create data collection tools, conduct interviews, and lead focus groups.

Leadership

I have undertaken a Trainee Editor Fellowship for the journal *Anaesthesia*, reviewing over 40 manuscripts, contributing to the Editorial Board and presenting at conferences. I have experience as a peer reviewer for the journals *Medical Education, BMJ Open Respiratory Research* and the *Journal of the Intensive Care Society*. I have collected data for national perioperative studies (SNAP-2, DREAMY and PQIP), and am listed as a study collaborator on 6 published manuscripts. I hold several leadership positions, including Chair of the Pan-London Perioperative Audit and Research Network, organising the Annual Scientific Meeting, selecting the projects supported by the network, providing research support, and delivering the annual Academic Anaesthesia Regional Training Day. I am the Equality, Diversity and Inclusion (EDI) Representative for the Faculty of Intensive Care Medicine resident doctor committee, setting up an EDI strategy and working on projects to tackle differential attainment. I am a Fellow of the South Asian Health Foundation, undertaking projects analysing health inequalities, which have been published in peer-reviewed journals and developing my understanding of the importance of EDI in research. Previously, I was the Academic Trainee Representative for the Imperial School of Anaesthesia, providing a point of contact for academic trainees and organising the North London Junior Anaesthetic Group conference.

Achievements

I have shown academic aptitude throughout my career, receiving a distinction for my dissertation and an Upper Second-Class Honours overall in my undergraduate degree, a Merit in Clinical Specialties and first place in a national prize while at medical school. During clinical training, I have completed the MRCP, a Diploma in Tropical Medicine & Hygiene, and the FRCA; been awarded several competitive travel bursaries to support an anaesthesia fellowship in Zambia; and received prizes for quality improvement projects. I have completed a National Medical Director's Clinical Leadership Fellowship at NHS England, observing the interaction between policymakers and academics. I contributed to policy documents (e.g., Evidence Based Interventions Programme), worked on national projects and collaborated with professionals from across disciplines. I developed interests in analysing health services and understanding how to navigate complex regulatory and commissioning processes.

Impact of Fellowship

While still at an early stage of my academic development, I believe that my track record of publications, collaboration, and skill attainment provides a strong foundation for the next step of research training. I am highly motivated to become an expert in the field of IMV, using mixed methods to CDSS using explainable machine learning that deliver personalised ventilation. This prestigious NIHR Doctoral Fellowship will act as a springboard for a career as a Clinician Scientist, leading my own research group and attaining a Chair in anaesthesia and critical care.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
I applied to the NIHR Doctoral Fellowship last year (Funding Round DF R11) with a project entitled "Machine learning driven visualisation of weaning from invasive mechanical ventilation" (reference number NIHR304745), and though I was invited to interview, my application was not successful at this stage. I am grateful to the reviewers for their comments and believe that addressing them has led to significant improvements to my proposal.

The reviewers identified me as a strong candidate but found 3 major issues with the application:

- Methodology
- PPIE involvement and clarity of the Plain English Summary
- The training plan required further detail

Methodology

I have worked in detail with my supervisory team, bringing on additional expertise through a collaboration with Dr Tim Stephens, an expert in qualitative methodology, Dr Sonali Parbhoo, a machine learning expert, statistician support, and the Imperial-NIHR Research Support Service (RSS). Work Package 1 now has a focus on dataset curation, identifying a core set of variables for ventilation analyses that can be applied to different datasets. Work Package 2 now has clearly described qualitative methods, has identified sites for recruitment, includes a Dephi to identify core parameters required for ventilation management, and includes co-design, which will better answer the research objectives. Work Package 3 now focuses on a clearer, measurable objective, assessing cognitive load using physiological signals, has stipulated how subjects will be recruited, and has a power calculation to inform sample size.

PPIE

I have undertaken additional activities, including seeking further input from the Patient Family and Advisory Group (PFAG) and presenting the project to parents of children admitted to the paediatric intensive care unit, 2 of whom have agreed to join the PFAG. I have brought on a public member co-applicant for the project, Ms Joanne Lloyd who will ensure the patient voice remains heard throughout and who will assist with co-design workshop delivery and analysis, as well as co-presentation of results.

Their input, and that of the RSS, has been instrumental in identifying ventilation outcomes that are important to patients, improving PPIE involvement in the project, and amending the Plain English Summary. The feedback of the reviewers was key here; all unsuitable words have been removed, sentence length shortened, subheadings added and the Flesch-Kincaid Grade Level on my Word processing software is now 7.2.

Finally, following feedback from the PFAG and reviewer suggestions, I have costed more regular PFAG meetings, scheduled PPIE involvement throughout the project, and costed preparation time and internet access costs for the PFAG.

Training & Development Plan

Having previously focused much of my training on machine learning methodology, I have now added external courses on qualitative methodology, implementation science, co-design and data visualisation. This will contribute to a much broader training experience.

Additionally, several minor issues were identified, including the title of the project, which is now more focused and easier to understand.

I have submitted a PhD application with Imperial (awaiting formal registration), and intend to start in February 2025. I have therefore costed my salary for 28m.

Contextual Factors

Please use this section to detail any factors that may have impacted the development of your research career or outputs from your research career to date. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research activities (including research outputs), due to disability or to caring responsibilities. This may include physical or mental health related impairments or conditions, which have impacted your opportunities to undertake research activities
- Reduced opportunities for career support e.g., mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, contextual factors should be significant, and relevant. NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with the General Data and Protection Regulations (GDPR).
N/A

4. Plain English Summary of Research

Aim

Improve the care of patients in intensive care who need help with their breathing.

Background

Every year, 70,000 patients in intensive care need help with breathing. This is delivered through a machine called a ventilator. To use a ventilator properly, clinicians must understand the patient and their disease. A clinical decision support system (CDSS) is a type of computer software that helps clinicians make better decisions. It matches patient and disease features to a database. Treatment options are then suggested to clinicians. The NHS Long Term Plan has suggested CDSS should be used to help care for patients.

Deciding when to reduce ventilator support is important. Leaving patients on a ventilator too long can cause lung injury. However, patients must be strong enough to breathe on their own. Previous research has shown differences in clinical practice when reducing ventilator support. There is mixed success and no agreement on best practice. This suggests CDSS could improve ventilator care.

Our national survey showed CDSS are not used for ventilator management. This is because there is limited ready-to-use computer software. The survey also showed clinicians do not trust CDSS. This is because their decisions are rarely explained.

Addressing the problem

This research aims to address these two problems.

1. Use computer-based methods to find patterns in patient data. This means studying lots of information from many patients. This method is called machine learning (ML).
2. Create new ways to show the data using visual tools. This will help people understand, trust and use CDSS better.

Design and methods

This research will be done in three steps.

1. Understanding the patient. We will use ML to predict when to reduce ventilator support. Predictions will cover the patient's whole stay in the hospital. They will show the key reasons for each prediction. We will check how well the models work and make sure they are fair for all groups. This includes those less often studied.
2. Designing new ways to show predictions using visual tools. Clinicians who work in intensive care will help create these tools, so they are helpful. The tools will show predictions and explain the important reasons behind them. This will be shown on an interactive computer system.
3. Checking the impact on care. We will test whether these tools make decisions harder or easier by comparing them to normal practice. We will do this through simulation testing. Our findings will guide future work. This could include testing CDSS in intensive care.

Patient and public involvement

This project has involved patients and the public from the start. This has shown this research can make a big difference in four ways:

1. Make clinical care more consistent.
2. Improve patient safety.
3. Help clinicians share information with patients and families using visual tools.
4. Help the NHS by shortening hospital stays and lowering costs.

This work focuses on two important areas from the James Lind Alliance. These have been chosen through discussion with patients.

Sharing our findings

I will share our findings through local scientific meetings and conferences. Results will be published in academic journals. We will use Imperial College London and collaborator networks to promote our work. This will be seen by industry, other research groups, and stakeholders. I have the support of a patient panel to help plan and share our work with the public. This will be supported by the Imperial College Science Media Team and public activities and events.

5. Scientific Abstract

Research question

Can explainable machine learning derive trajectories for successful and unsuccessful weaning from invasive mechanical ventilation? Can data visualisation of such explanations aid clinical management?

Background

Invasive mechanical ventilation (IMV) is an important intervention in intensive care, delivered to 70,000 patients annually in the UK. Ventilatory weaning is the process of reducing and removing ventilatory support and shows variable success, with significant heterogeneity in practice. Clinical decision support systems (CDSS) are software that integrate patient data with a computerised knowledge base to provide treatment suggestions. They can support delivery of personalised ventilation to optimise weaning. However, few CDSS exist due to a lack of readily deployable software, and in feasibility trials, advice adherence is low. A national survey found a key barrier to implementation is that CDSS are “black box”, failing to demonstrate rationale for their decisions.

This project will apply novel machine learning (ML) approaches within large longitudinal clinical datasets to delineate weaning trajectories in adults and children. It aims to test whether explainable artificial intelligence (XAI) can provide understandable and transparent explanations of decisions and actions to human, through novel data visualisation techniques.

Aims and objectives

1. Iteratively develop and validate XAI-based models to delineate subgroups based on physiological trajectories.
2. Develop and optimise data visualisation approaches to improve model transparency for clinical application.
3. Validate data visualisation approach using simulation.

Methods

I will use a novel ML approach to generate predictions, using three neural network architectures (Temporal Fusion Transformer (TFT); Temporal Convolutional Networks (TCN); Long-Short Term Memory Nets (LSTM)) designed for processing sequential, time-series data to provide multi-horizon forecasts. This architecture will be applied to large longitudinal clinical datasets, totalling over 100,000 patients. An iterative, expert-in-the-loop approach will interrogate the models.

Clinician and patient focus groups and co-design workshops will be conducted to understand stakeholder requirements and develop novel data visualisations utilising a web front-end to display the results, displayed through the Data Observatory at the Imperial Data Science Institute. These will be optimised for clinical use through iterative co-design cycles.

Influence on cognitive load and decision confidence will be assessed with a human-artificial intelligence (AI) interaction vignette study using a modified, between-subjects design. Think-aloud techniques will be used to understand rationale for decisions.

Timelines for delivery

I expect to develop and validate XAI-based models in months 1-12; conduct focus and co-design workshops to develop and optimise visualisations in months 12-27; and conduct simulations in months 27-33. I will write up my thesis in the final 6 months of the Fellowship.

Anticipated impact and dissemination

This research will have significant impact. Patients will benefit through receipt of efficacious, evidence-based therapy, clinicians through reduced heterogeneity in practice, and the NHS through optimisation of patient time on the ventilator, reducing length and cost of admission.

Our findings will be disseminated through local scientific meetings, conferences, and publications. I will utilise established collaborator networks of key stakeholders to identify shared learning. I have convened a patient and family advisory group who will assist with co-presenting our message for public dissemination.

6. Detailed Research Plan

1. What is the problem being addressed?

Clinical Need:

Invasive mechanical ventilation (IMV) is a key supportive therapy for patients on intensive care (ICU)[1]. In the UK, over 54,000 adults and 13,000 children required IMV in 2021[2,3]. Ventilation decision-making is complex, involving integration of clinical data from many sources. While advances have been made in the field[4–6], there is significant heterogeneity in ventilation practice[7]. The goal is to deliver personalised care, tailored to the individual[8].

Health Technology Solutions:

Clinical decision support systems (CDSS) for IMV aim to enable personalised care[9]. CDSS match patient characteristics to a computerised knowledge base and generate patient-specific recommendations; they are divided into open-loop (clinician input required to enact suggestion) or closed-loop (operate ventilator independently)[10]. Currently available CDSS for IMV are predominantly based on physiological models[9,11] or rule-based algorithms[12,13].

Ventilatory weaning is the process of reducing and removing ventilatory support as patients improve. Our narrative review suggests weaning is where CDSS may be most valuable, as it is multidisciplinary, there is significant heterogeneity, and success rate is low[14]. It is a critical phase that every ventilated patient must undergo; early liberation reduces the risk of nosocomial pneumonia, ventilator-induced lung injury[15], and prolonged admission[15,16], while premature discontinuation can cause fatigue, impaired gas exchange and reintubation[17,18].

A feasibility study testing an open-loop CDSS from our group showed bedside staff rarely followed CDSS advice[19]. A national survey conducted in response to this implementation gap found clinicians perceive the “black box” aspect of CDSS particularly troubling, and limited devices that can be implemented in the NHS[20–22].

Enhanced understanding and adoption through data visualisation and explainable AI (XAI):

Machine learning (ML) can be employed to analyse patient data and identify patterns that predict weaning outcomes, though previous studies have not employed time-series data[23–25]. Multi-horizon forecasting (MHF) is the prediction of variables over several points in time. In ventilatory weaning, a single-horizon forecast predicts settings for the next hour, whereas MHF predicts parameters over several days or weeks, enabling continuous optimisation of care.

Explainable artificial intelligence (XAI) is the development of ML models that provide interpretable explanations for their predictions. In the context of MHF, explainability is crucial for understanding rationale. We believe data visualisation can deliver on the implementation of explainable MHF. Visualisation has been used to illustrate patterns in data, communicate findings to expert and lay audiences[27], and explore dynamics of Bitcoin transactions[28], online discussion[29] and bio-surveillance[30]. Data visualisation in ICU has been limited to live dashboards and handover of care[31–33]. Showcasing patient ventilatory trajectory to augment decision-making is novel.

Our group has applied XAI to illustrate feature importance; analysis of SHapley Additive exPlanations (SHAP) values derived from time-series models provided insights into the features impacting a model assessing patient trajectory in COVID-19[7]; alluvial plots demonstrated how physiological trajectory linked to patient outcome[7].

Hypotheses:

1. ML models can identify weaning trajectories through analysis of longitudinal clinical data.
2. Data visualisation can display trajectories to augment clinical decision-making.

This project aims to improve understanding of weaning from IMV using ML, creating MHF models that use data visualisation to improve model transparency. This will reduce heterogeneity in weaning practice, augment delivery of personalised ventilatory care, and increase acceptability of CDSS for clinicians.

2. Why is this research important in terms of improving the health and/ or the wellbeing of the public and/ or to patients/ service users, carers and health and care services?

This project tackles an important problem that is recognised by clinicians, policymakers, and patients.

Ventilated patients represent nearly 40% of all adult[2] and 60% of all paediatric admissions to critical care[3]. The proportion of patients receiving IMV in ICU is increasing[34], and IMV remains the main reason for patients staying on ICU[35,36]. Mortality from conditions such as acute respiratory distress syndrome (ARDS)[1] and ventilator associated pneumonia[37] are high, and for patients who undergo prolonged ventilation, recovery is complex[38]. Delivering personalised ventilation strategies that optimise time on the ventilator is key[8].

Moreover, an increasingly multidisciplinary workforce require tools to assist with delivering safe care[39]. The NHS Long Term Plan has called for the development of decision support tools to help clinicians apply best practice[40], and the Topol Review recommended artificial intelligence should be used to deliver personalised care[41]. Despite this need, our prior work showed that CDSS for IMV are rarely used in the UK[19].

Our public and patient representatives, who I have engaged through the charity ICUSteps (https://icusteps.org/), VOICE (https://www.voice-global.org), the Research Partner's Group, the Research Support Service and the Patient Experience Research Centre at Imperial, include members with lived experience of both adult and paediatric ventilation. I have presented to these groups several times, and the feedback has been very positive, with respondents agreeing this was a very important research topic that could have a substantial impact for patients. The feedback has been instrumental in reviewing the lay summary, the project proposal, and identifying outcomes that are important to ventilated patients: these include reducing time on the ventilator and improving comfort by weaning at the most opportune time, both of which this research will address. Public participants also identified that data visualisations could be used as a communication aide for conversations with family members and carers, illustrating complex data to improve understanding and achieve treatment concordance. They highlighted the importance of interrogating models for bias, maintaining confidentiality and provided guidance for convening a Patient and Family Advisory Group (PFAG) to ensure this work remains relevant to patients and the public. We will meet every four months to discuss our findings and guide the project. The PFAG will craft our message for public dissemination and co-present our results at conferences. I have involved a public member co-applicant with experience of ventilation research, who has been instrumental in fine-tuning the lay summary, and will support the project through participation in co-design workshops, data analysis and co-presentation at conferences.

This project addresses two high research priority areas of the James Lind Alliance, arrived at through consultation with patients and carers ("What is the best way of ensuring new knowledge and the latest defined standards are introduced into clinical practice in a timely and effective way?" and "How can we predict who will benefit from intensive care before admission and during treatment in the ICU?").

3. Review of existing evidence – how does the existing literature support this proposal?

We performed a narrative review, published in the *British Journal of Anaesthesia*, aiming to understand factors that influence clinical practice and decision-making for IMV, both with and without CDSS, and facilitators and barriers to implementation. The database search identified 2,616 studies, of which 30 met the inclusion criteria.

Role allocation was well described, with nurses most likely to lead decisions around weaning readiness and failure; and physicians on selection of weaning method and extubation readiness. Interprofessional collaborative decision-making was considered important, and is similar between adult and paediatric critical care, but more common in western Europe, explained by differences in role allocation, training requirements and cultural factors. Nursing autonomy and influence on clinical decisions was dependent on nurse: patient ratio, more likely when 1:1. Unit culture and reduced perception of clinical reprimand leads to greater autonomy in decision-making by junior staff[42–51]. However, it is unclear whether this varies with units of different leadership models, size, staff experience and composition.

There was no information on clinical workflow and no studies defined process measures for ventilation, outside of morbidity, mortality, and weaning success (which itself has several definitions). Arnal and colleagues assessment of INTELLiVENT-ASV, a closed loop CDSS, showed the potential of these devices to augment workflow: this CDSS significantly reduced manual ventilator setting changes[52].

Weaning of IMV was explored in more detail. Nurse-led weaning is broadly grouped into patient, nurse, and system factors[49,53–61]. Physician-led weaning is poorly understood, with significant heterogeneity in practice, and no evidence exploring impact of clinician specialty, experience, or level of training. Physicians, while using a combination of subjective and objective information, are more likely to draw on "gestalt"[57,62–64]. The success of physician-led weaning is low, less than 50% in some studies[14]. Clinical vignette studies have shown significant heterogeneity in the data required to assess patients, ventilation practice, and limited between-clinician agreement[65,66]. The picture is complicated by variability in critical care definitions and treatment criteria[67].

No studies explored decision-making while using CDSS. Facilitators and barriers to implementation of care bundles in ICU are broadly grouped into those affecting the clinician, intervention and organisation[20–22,68].

One study described facilitators to implementation of CDSS; these included integration with existing electronic health records and computer systems, robust monitoring and evaluation, and adequate technical and clinical support[20]. Other studies suggested unstructured approaches to the implementation of care bundles[68,69]. No studies documented use of implementation frameworks to guide transfer to clinical practice.

Our national survey highlighted that CDSS for IMV are rarely used in the UK. Only two CDSS were in use, neither of which were used often or effectively, because decisions were not explainable ("black box") and the evidence-base was perceived insufficient to support implementation. However, respondents felt CDSS could improve clinical decision-making, patient safety and outcomes, provide data for audit and research, and facilitate junior staff training. Barriers to implementation including resistance to changing practice, deskilling of staff and device complexity.

Patient subgroups have been identified in ventilated patients[70], with the potential for therapeutic applications shown in several retrospective analyses[71,72]. Physiological optimisation based on these disease state is effective[73–75]; for example, in severe ARDS, higher PEEP was associated with better survival[76], while ventilator strategy misaligned to lung morphology was associated with worse survival[77]. Thus, optimising ventilator management through improving understanding disease state is key to improving patient management.

Longitudinal ventilation data from patients with COVID-19 induced ARDS identified clinical trajectories that were highly discriminatory for patient prognosis[78]; this approach has also demonstrated the trajectory of ARDS patients[79]. Further work is required to identify whether longitudinal physiological data can further define groups of ventilated patients.

Our review of the literature and background work has demonstrated poor understanding of weaning. While several CDSS for IMV have been designed[9,11,12,13,80,81], explainability of these models is limited and hinders implementation. Promising work has identified patient trajectories that are predictive of prognosis. We believe that this represents a significant opportunity for further research.

4. What is the research question/ aims and objectives?

Research question

Can explainable machine learning derive trajectories for successful and unsuccessful weaning from invasive mechanical ventilation? Can data visualisation of such explanations aid clinical management?

Objectives:

1. Iteratively develop and validate XAI-based models to delineate subgroups based on physiological trajectories.
2. Develop and optimise data visualisation approaches to improve model transparency for clinical application.
3. Validate data visualisation approach using simulation.

5. Project plan

Objective 1:

- To identify subgroups based on patient physiology and response to ventilatory interventions.
- To identify features that define data-driven subgroups.
- To define impact of features with MHF.

All ventilated patients will be eligible for inclusion. Using a-priori heuristics defining weaning as the first discrepancy between set ventilator and respiratory rate, previously piloted by our laboratory group, I will stratify patients ventilated for < 24 hours, between 1-7, 8-21, and > 21 days. Weaning outcome will be based on core outcome set for critical care ventilation trials (success – extubated for > 48h, free from all tubes, endotracheal tubes and tracheostomy; failure – reintubation or death following extubation)[82].

Patients will be drawn from databases with data sharing agreements in place: MIMIC-IV (40,000 patients), Guy's and St Thomas' (GSTT, 20,000 patients), Royal Brompton & Harefield (RBHT, 20,000 patients) and Imperial College Healthcare NHS Trusts (ICHT, 20,000 patients), and aggregated data from three paediatric intensive care units (PICUs) across ICHT, RBHT and Great Ormond Street NHS Trusts (15,000 patients). Traditional power

calculations are not required for ML analysis, as there is no specific hypothesis being tested; thus, size of subgroups and the effect size are not known a-priori. MIMIC-IV is hosted on Google Cloud platform as a Structured Query Language (SQL) database; GSTT, RBHT and ICHT data are stored in Trusted Research Environments on secure NHS Servers as SQL databases; and the paediatric ventilation data is stored in the Big Data and Analytics Unit Secure Environment (BDAU SE) as an SQL database. NHS data will be anonymised (MIMIC-IV is already anonymised), and all data will be extracted into CSV files.

First, I will curate datasets. Using clinical domain expertise with supervisors Dr Patel and Dr Ramnarayan, I will identify variables that are present across all datasets, with parameters likely to include those in the ventilatory, physiological, and laboratory domains, and patient demographic and admission information. Thus, our analyses will be performed on clinically universal, rather than site-specific, features. Data pre-processing and exploratory data analysis will be performed to understand correlation, collinearity, missingness, and outliers. Variables will be interrogated to ensure data is consistent with its label, has appropriate units, and temporal relationships are maintained. Where possible, variables will be standardised and normalised, and descriptive statistics and simple visualisations e.g. histograms, will improve understanding. Where significant data missingness is identified, I will use imputation methods, such as parameter-specific time limited forward imputation for time-series data. Noise and outliers will be identified with domain expertise, removing impossible values (e.g. negative physiological values). I will use simple tests of association, such as regression analyses, to identify signals in the data.

Initially, each dataset will be analysed individually, to account for variation in populations. Dataset will be divided into training (80%) and test (20%) sets. Using signals identified from regression analyses, I will assess the performance of ML architectures such as Random Forest and Support Vector Machine. I will then assess the Temporal Fusion Transformer (TFT), an attention-based deep neural network architecture for MHF, which provides robust forecast performance, is explainable, and identifies patterns in time series data. Knowing which factors (feature importance) and historical data points (temporal attention) contribute to predictions will demonstrate rationale. Feature importance will be demonstrated using SHAP dependency plots, prompting clinicians to consider different data dependent on point of assessment[7]. Hence, I will assess whether, and how, ventilatory interventions affect predictions. I will assess for commonalities in weaning trajectories and feature importance across datasets; if appropriate, I will combine datasets to create a universal model that can be applied to all adult populations. Analysis of paediatric and adult datasets will remain separate due to differences in physiology and ventilation practice between these two populations.

I will promote human-AI collaboration through an iterative, expert-in-the-loop approach. If the TFT model does not perform as expected, I will combine Temporal Convolutional Networks and Long-Short Term Memory Nets, convolutional and recurrent neural networks, respectively. The final design will minimise bias while maximising performance and avoiding over-fitting.

Model performance will be assessed in terms of accuracy (i.e. ability to predict outcomes across different weaning timeframes), sensitivity, specificity, F1 score (weighted according to class weights), receiver operating characteristic curve and area under the receiver operating characteristic curve (AUC). I will perform 5-fold cross-validation and select the model with the highest AUC. Models will be assessed for bias, calibration in under-represented groups, and I will be vigilant for leak of demographic data. I will assess algorithms against international standards for ethical ML and ensure reporting of results conform to SAGER guidelines and NIHR-INCLUDE Ethnicity Framework. I will collaborate with the DSI and supervisor Dr Serban for this analysis.

Objective 2:

- To understand and incorporate key information required for ventilatory weaning and decision-making into data visualisations
- To define core variables required for ventilatory weaning
- To co-design, with clinician users, data visualisation that will aid clinical decision-making
- To co-design, with clinician users and patients, a user interface that enables clinical interpretation of visualisations, such as displays for weaning trajectories and key ventilation parameters

To understand key information required for ventilatory weaning, I will conduct focus groups, recruiting 20 clinicians from five ICUs of varying specialisation: a respiratory failure centre (Royal Brompton Hospital), PICU (St Mary's Hospital), and general adult ICUs (Royal London Hospital, Hammersmith Hospital, Kings College Hospital). Agreements are in place with all, with a named contact at each. I will purposively sample to ensure a mix of sex, ethnicity, professional background (medical, nursing, advance critical care practitioners and physiotherapy) and clinical experience. Sampling from different units will allow an assessment of unit-based practice. While we acknowledge consensus may never be achieved, we believe 20 clinicians represents an adequate sample based on prior work by our group. Focus groups will be led by an experienced facilitator, collaborator Dr Tim Stephens, who will ensure equal participation and a flattened hierarchy. Each focus group will last 90 minutes and will include 6-7 participants. Questions will be designed with Dr Stephens and supervisor Dr Ni and will be piloted before use. They will be video recorded, and field notes will be taken contemporaneously. Videos will be transcribed, coded and undergo thematic analysis based on Braun and Clarke's reflexive thematic analysis[83] using a hybrid of

inductive and deductive approaches. Data analysis will be conducted by myself, Dr Stephens and Dr Ni, with identified codes and themes compared for concordance, identifying key variables and decision-making rationale.

Next, I will conduct a Delphi to define a core set of variables required for ventilatory weaning, with UK experts identified by supervisor Dr Patel, Dr Padmanabhan, and collaborator Dr Camparota. Rating rounds will be conducted remotely and anonymously, using Qualtrics software. Participants will be asked to assess variables identified from the focus groups, and to determine which are considered essential for ventilatory weaning. Respondents will be asked to rate each on a 4-point Likert scale (definitely include, probably include, probably exclude, definitely exclude). A threshold of $\geq$ 75% of panellists agreeing with a variable will warrant inclusion. Variables that rate as either “definitely include” or “probably include” by > 50%, but do not meet the inclusion threshold, will be retained for the next round, while variables that don’t will be excluded. Three rounds will be conducted, after which variables that meet the inclusion criteria will be put forward as a “strong recommendation”. Participants will be able to suggest alternative variables.

Using these findings, we will design visualisations (e.g. time series plots, scenario analysis charts, heat maps, temporal attention plots, interactive dashboards) to display different aspects of the model. Visualisations will be showcased through the Data Observatory at the DSI.

I will conduct two rounds of co-design[84] workshops with the clinicians from the focus groups, our PPIE co-applicant, and 3 representatives from the PFAG, to iteratively optimise visualisations. The first workshops will seek to understand which visualisations are beneficial for clinicians and patients. Workshops will last four hours and include 8 participants. Dr Stephens will facilitate to ensure equal participation and a flattened hierarchy. Workshops will follow published guidance[85,86], and use co-design techniques to facilitate discussion and elicit design input. These will include storyboard techniques (use of scenarios to illustrate how the tool might work in practice) to test feasibility, and display of visualisations to test acceptability. Co-design workshops will be designed with the assistance of the Imperial Healthtech Research Centre (HRC) and collaborator Dr Stephens, overseen by supervisor Dr Ni. Activities and questions will be piloted before use.

Following this, I will create a user interface (UI) using a system architecture that incorporates and processes the data, and a web front-end to display the results, built around free-lock pipelines with reusable components that transform data into a desirable format[27,30]. Tasks will be run in the background due to required data processing volume. Processed data will output to a front-end data store (PostgreSQL, an open-source relational database management system); the front-end UI will run as HTML5 and React.js JavaScript app. The UI will enable interaction and filtering of data. It will undergo the second round of co-design using validated tools[87], testing feasibility and optimising for use. “Think-aloud” techniques, where users talk about what they are doing while interacting with the UI, will be used to understand decision-making[88]. The optimised UI will be beta-tested with 10 clinicians from the co-design groups, using the System Usability Scale and think aloud techniques to understand usability in vivo.

This patient and clinician user informed co-design approach will establish a deeper understanding of acceptability of the intervention, usability needs for the interface, and implementation requirements. This will involve our NHS partners and clinical engineering at Aalborg University to help with UI design.

Objective 3:

- To assess how XAI using visualisations affect cognitive load.
- To assess how XAI using visualisations affect decision confidence.
- To describe decision-making rationale while weaning IMV.

I will conduct a human-AI interaction vignette study using a modified, between-subjects design. Participating clinicians will require at least 5 years of clinical experience on ICU and regularly adjust the ventilator in clinical practice. We will purposively recruit to ensure a mix of professional background and experience, using links established with the Intensive Care Society, who hosted our national survey, to advertise the opportunity and reach clinicians across the UK.

Each subject will perform 16 trials, with the first four used as a training period and thus identical for each participant. In the subsequent 12 trials (the main experiment), subjects will be asked whether they would commence, continue or stop weaning, and to determine ventilator mode and settings for the next hour, while describing their decision-making rationale. Trials will be selected from real-world patient data from the analyses in Work Package 1, by ventilation experts Dr Patel and Dr Ramnarayan, and will represent all four ventilation time stratifications, ensuring a representative mix of complexity.

I will use a multifactorial experimental design with three arms, where each clinician will experience all three arms on different patient subsets. The three arms are: baseline with no additional AI information (baseline); additional AI CDSS information (AI); CDSS with XAI decision using data visualisation (XAI).

The primary outcome will be difference in cognitive load between the three arms, measured through physiological signals from the Empatica EmbracePlus wristband. These include electro-dermal activity (EDA, or skin conductance levels), heart rate variability and skin temperature (SKT)[89]. EDA measures electrical conductance properties of the skin, influenced by sweating, and is used to detect changes in sympathetic activity caused by cognitive processing[90]. EDA has two components, slowly changing skin conductance level (SCL) and rapidly changing skin conductance response (SCR, peak amplitude 1-4 seconds after stimulus). Higher heart rate and SKT are also related to increased sympathetic stimulation and used as a measure of cognitive load. Cognitive workload experienced by each participant will be measured by comparing physiological signals with a baseline state, where the participant is not performing any task. Data will be normalised, pre-processed and segmented into 5-second chunks for feature extraction (accounting for peak SCR), and statistical metrics including central tendency (e.g. mean, median) and dispersion (e.g. range, standard deviation) will be calculated.

Repeated-measures analysis of variance (ANOVA) will be performed to compare mean differences of physiological signals across the three arms. A power calculation has shown that a sample of 102 clinicians will provide the experiment with a 90% power to detect a difference in mean physiological signals, assuming an effect size of 0.25 and a significance level of 0.05. Participants will fill in the NASA-Task Load Index (NASA-TLX)[90] after participation in each arm of the study, measuring subjective cognitive load, and correlation analysis will be performed.

Clinicians will be asked to engage in think-aloud techniques, expressing rationale for their decisions and decision confidence on a numerical scale. Simulations will be audio recorded, which will be transcribed, coded and undergo thematic analysis based on Braun and Clarke's reflexive thematic analysis[83] to identify decision-making rationale.

6. Dissemination, outputs and anticipated impact

Our findings will be disseminated through local scientific meetings, national and international conferences (e.g. the American Thoracic Society Annual Congress). My supervisory team and the wider research group will support me to publish in peer-reviewed, high-impact journals. I will utilise established NIHR HRC and Imperial College networks of industry, clinician, research, public and patients to identify areas for shared learning (e.g. Severe Acute Respiratory Failure Network and Imperial AI Network). I have established collaborations with Aalborg University, the VentilaGroup study team in Spain, and with UK groups, including two specialist long-term weaning centres. We have received assistance from the Intensive Care Society, the Paediatric Critical Care Society and Critical Care Research Group for our national survey and will continue to use these networks to disseminate our work. Our research group at Imperial has also collaborated with several international groups. I will ensure reproducibility by sharing code through repositories such as Zenodo and in the supplementary materials of accepted publications.

We will liaise with our PFAG and co-applicant to contribute to written material for public dissemination, and to co-present our findings at conferences. We will utilise the Imperial College Science Media Team to publicise our work, utilise their public outreach activities, and disseminate through groups such as the British Lung Foundation.

This research will have significant impact for patients, clinicians, and the NHS. Patients will benefit through receipt of efficacious, evidence-based therapy, improving safety, reducing complications and time on the ventilator. Clinicians will benefit through improved understanding of ventilatory weaning, reducing heterogeneity in practice, and improving communication with patients and families. The NHS will benefit through optimisation of patient time on the ventilator, reducing length of stay and cost of care.

7. Project management

This Fellowship will be under the supervision of Dr Brijesh Patel, Clinical Senior Lecturer in Cardiothoracic Critical Care at Imperial College London, with whom I will have weekly meetings to monitor progress, and who will maintain high-level oversight.

I will be co-supervised by Dr Padmanabhan Ramnarayan, Professor of Paediatric Critical Care at Imperial, who will provide guidance on paediatric ventilation and expertise in clinical informatics and AI; Dr Melody Ni, methodologist and Deputy Director of the NIHR HRC at Imperial, supervising methodological and implementation aspects, providing expertise in translating technology into clinical practice and links to NIHR centres, including HRCs; and Dr Ovidiu Serban, Research Fellow at the Imperial DSI, who has expertise in ML and large-scale data visualisation and will provide links to the Data Observatory.

Additionally, I will be supported by collaborators including Professor Steve Rees, Professor of Respiratory Critical Care Technology at Aalborg University, with expertise in CDSS and biomedical engineering; Dr Tim Stephens, Clinical Lecture and Implementation Scientist at Queen Mary University of London, with expertise in qualitative research and methodology; and NHS partners, including Dr TC Aw, Associate Chief Clinical Information Officer at the Royal Brompton Hospital, an expert in real-world clinical data development; and Professor Luigi Camporota at Guy's & St Thomas' NHS Trust, who has extensive clinical and academic expertise in IMV.

8. Ethics

Several issues relating to AI in healthcare apply to this project. These include patient confidentiality, data security, data transparency and accountability.

Data for this retrospective observational study has either been anonymised and stored on a Google Cloud platform (MIMIC-IV), which meets data protection security standards; is pseudo-anonymised, stored on secure servers with access granted by following organisational procedures, and will be anonymised prior to analysis (GSTT, RBHT and ICHT databases); or has been granted ethical approval for use in ventilation analyses (paediatric database, IRAS 323325). I will apply to the Health Research Authority for use of NHS data, but assessment with their online tool suggests Research Ethics Committee review is likely unnecessary. Patient consent will not be required as all data will be anonymised. Data sharing agreements are in place for all databases, which have undergone processing for quality standards. ML trains on the whole patient cohort, so risk of identification is further reduced. Data at Imperial is stored on a cloud-based system that meets the College's standards for data security and is compliant with UK and EU Data Protection Laws. To minimise risk of unauthorised access, Jupyter notebooks will stay in the Imperial laboratory. All code will be maintained in Jupyter notebooks, stored in the online data repository, Zenodo, and shared through publications, improving transparency and allowing replication of our analyses.

Aware of the potential for AI to exacerbate inequalities by replicating established social and racial biases, I will review for bias and assess for calibration in under-represented groups against international standards for ethical ML. Reporting of results will conform to SAGER guidelines and the NIHR-INCLUDE Ethnicity Framework. I will purposively sample for the focus groups, co-design workshops and simulation to ensure a distribution of age, sex, and ethnicity that reflects the population.

Alongside presenting to and seeking feedback from several patient and public panels, I have convened a 12-member PFAG and identified a public co-applicant, with lived experience of IMV to advise on study design, data analysis, ethical issues, and dissemination of results. Their feedback has been instrumental in designing this proposal.

9. Success criteria

Success will be judged by dissemination of outputs detailed in the project plan, and completion of the Doctorate. Additionally, the success of each objective will be judged as follows:

1. **Objective 1**
- Identify and externally validate weaning trajectories using time-series ML.
- Quantify feature importance and variation with time.
- Build a predictive classification model that can generate MHF.
- Demonstrate how clinical interventions affect predictions.
- Review model for bias and assess for calibration in under-represented groups.
2. **Objective 2**
- Define key parameters required for ventilatory weaning and decision-making.
- Develop visualisations that display data.
- Develop a UI that enables interaction with the data.
- Optimise for clinical application through co-design.
3. **Objective 3**
- Assess impact of XAI with visualisation on cognitive load and decision confidence.

10. Risk mitigation

A major issue with retrospective observational data relates to data quality. MIMIC-IV is a high granularity database that is accepted as a reliable data source, with good documentation and harmonisation between datasets, and over 2,000 publications to date. All code used for generating tables for analysis is publicly available (https://github.com/MIT-LCP). The ICHT, RBHT, GSTT and paediatric critical care databases have undergone processing for quality standards; database curation for the NHS databases has been costed into this application.

The application of three neural network architectures (Temporal Fusion Transformer, Temporal Convolutional Networks and Long-Short Term Memory Nets) to identify weaning trajectories will mitigate the risk of model failure or poor performance. The involvement of supervisors and collaborators from several disciplines will mitigate against failure to complete the research objectives.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

I have engaged members of the public through the charities ICUSteps (https://icusteps.org/) and VOICE (https://www.voice-global.org), and through Imperial College London, presenting at the Research Partner's Group (RPG) and the Patient Experience Research Centre (PERC), to those with lived experience of adult and paediatric ventilation. I have received feedback from the Research Support Service (RSS), who consult members of the public. Finally, recruiting through VOICE and PERC, I have convened a 12-person Patient and Family Advisory Group (PFAG), and a co-applicant, Ms Joanne Lloyd, who will advise on the project.

Presentations to the RPG and PERC has involved 3 iterations of the project; I presented to 6 panel members at the RPG, and 2 panel members through PERC, who provided engaging feedback. All agreed this was a very important research topic which could significantly benefit for patients and the NHS. This was reflected in feedback from the RSS, which highlighted the potential to reduce variation in clinical practice.

The lay summary initially contained jargon and objectives were unclear; after the input of the RPG, PERC and the PFAG, it was considered to have captured the project's main themes and was easy to understand. The lay summary was also sent to members of the charity ICUSteps, and feedback was very positive, with one respondent commenting, "It's clear ***what*** the research wants to look at, why they are doing it and how they will do it. It is particularly encouraging to see the development of the need to understand the patient as an individual."

Following presentations to the RPG and PERC, themes to emerge from the discussion were:

- Potential for artificial intelligence to perpetuate bias.
- Data security and maintaining confidentiality.
- Importance of transparency when reporting results.
- Ventilation outcomes that are important to patients.
- Potential for visualisations to communicate complex information to patients and family members.
- Guidance on recruiting lay panel:
    - Importance of including panel members with lived experience of intensive care.
    - Ensuring diversity amongst panel members.
    - Ensuring appropriate safeguards in place.

This has led to amendment of this proposal, and following discussion, we allayed these concerns and recruited the PFAG.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

I have convened a PFAG to advise on the project. Feedback from the RPG was instrumental for reducing barriers to participation; where I had initially considered good written and spoken communication skills, and recruiting on a first come, first served basis, the RPG suggested this could led to a homogenous group and exclude certain voices. Subsequently, criteria for recruitment were that participants should have lived experience of intensive care, either as a patient or family member, should be in stable recovery of at least 3 months since ICU contact; and > 18 years old. The group we have purposively convened consists of 12 people, age range 34-74, 75% female, 25% non-white, from all over the UK, and with lived experience of adult and paediatric ventilation.

Their feedback has been instrumental in developing the proposal. We will meet at least four-monthly over video conference to discuss our findings and guide the project. The PFAG will also assist with crafting our message for public dissemination and contribute to written material. We will support PFAG members with development opportunities through PERC. One member of the group, Ms Joanne Lloyd, who has experience as a PPIE member for other mechanical ventilation studies, has agreed to join the study as a co-applicant. Her expertise will be invaluable in assisting with co-design workshops, co-presenting results at conferences, and ensuring the patient voice remains at the forefront.

Additionally, we will use the Imperial College Science Media Team to publicise our work, and disseminate through groups such as the British Lung Foundation and ICUSteps to reach a wider audience.

For PPIE training and support, I will seek support from PERC, including attending their 4-week Public Involvement Training course, and seek support from my supervisors and collaborators, all of whom have experience of involving patients in research. This proposal has also costed the time of the PFAG and co-applicant in accordance with the NIHR Centre for Engagement and Dissemination guidelines.

To evaluate PPIE involvement and capture its impact, I will maintain all patient involvement in the PERC log and regularly evaluate progress with the PFAG and my supervisors.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
N/A

8. Training & Development and Research Support

Proposed training and development programme

I am an early career researcher, highly motivated to become an expert in the field of invasive mechanical ventilation (IMV), using mixed methodology to develop clinical decision support tools using explainable artificial intelligence (XAI). I aim to use the opportunity of this prestigious NIHR Doctoral Fellowship as a springboard to launch a career as a Clinician Scientist, lead my own research group and attain a Chair in anaesthesia and critical care.

Supervision

In the short-term, I intend to complete a high-quality PhD under the guidance of Dr Brijesh Patel and Dr Padmanabhan Ramnarayan, physician scientists in adult and paediatric critical care and ventilatory support; Dr Melody Ni, decision analyst and methodologist with a background in translating innovation into clinical practice; and Dr Ovidiu Serban, senior data scientist with expertise in machine learning and largescale visualisation projects. The expertise in my supervisory team will empower me to deliver on this ambitious project while acquiring new research skills, disseminate my work through high-impact publications and develop into an independent researcher. Gaining Fellowship support from the NIHR will allow me to focus on these goals. Embedding within the Imperial Data Science Institute (DSI), I will learn how to apply novel time-series machine learning to understand weaning trajectories and use novel data visualisations to deliver on the goal of XAI. I will obtain support from the Imperial-NIHR Healthteach Research Centre (HRC) to learn, develop and assess optimal approaches to improve clinical implementation. This is a novel approach for producing XAI tools. I will draw on my background in data science, the experience of my supervisory team, and collaborators: machine learning (Dr Parbhoo at Imperial), ventilation (Dr Camporota at Kings College Hospital), qualitative research (Dr Stephens at Barts Health), and decision support development (Professor Rees at Aalborg University), to deliver on these goals.

Courses

I will attend local courses at Imperial College, accessed through the Clinical Academic Training Office, who provide academic and pastoral support for clinical PhD students, and host courses ranging from statistical analysis to academic writing. I will attend the Public Involvement Training course, run through the Patient Experience Research Centre at the start of my Fellowship.

To develop skills in qualitative methodology, implementation science and co-design, I have costed the “Qualitative Research Methods in Health” course at University College London, “An Introduction to Implementation Science” at Kings College London, and “Experience-Based Co-design”, delivered by the Point of Care Foundation. Additionally, following supervisory recommendation, I will attend the “Developing & Evaluating Complex Interventions” course, delivered online by the University of Glasgow. This will be important for understanding how complex interventions (clinical decision support for IMV) can be implemented in a complex environment (critical care).

To improve skills in machine learning and coding, I have costed annual membership for Coursera (www.coursera.org) and Datacamp (www.datacamp.com). For example, on Coursera, the “AI for Medicine” course includes modules on time to event modelling, useful for planned analyses. Courses on Datacamp will be key for improving coding skills. I will attend the “Mathematics for Machine Learning” from the MSc in Computing at Imperial, which is included as part of Coursera membership. Imperial College runs courses as part of the UKRI AI for Healthcare Centre for Doctoral Training. I will utilise these to address gaps in my knowledge.

As a basis for using visualisations to improve understanding of complex data, I will attend the “Data Visualisation and Visual Analytics” course at the Alan Turing Institute, which is delivered online.

International collaboration

I will visit Aalborg University to mature a collaboration with Professor Steve Rees and his team. Professor Rees is an expert in biomedical engineering, decision support systems and physiological modelling. During an introductory visit in August 2022, we worked on a preliminary analysis using physiological data to subgroup patients weaning from the ventilator. We have subsequently worked on a narrative review assessing clinical practice and decision-making in patients requiring IMV (published in the *British Journal of Anaesthesia*).

This visit, costed into the PhD, will be key for developing skills in data analysis, machine learning, and decision support, enriching my development and that of our critical care research group in London, while fostering ongoing academic relationships.

The future

On completion of the PhD, I will apply for a post-doctoral fellowship (e.g., NIHR Clinical Lectureship) to develop as an independent researcher with a formal arrangement for postdoctoral training. Clinically, I continue Anaesthesia and Intensive Care Medicine registrar training, where I will work towards my Certificate of Completion of Training as a dual specialist. I will continue to seek guidance from supervisors and mentors.

Long-term, I am motivated to follow a clinical academic track. I intend to obtain a clinical academic faculty position in a university and associated hospital, facilitating my research goals. At present, my academic interests are focused on using machine learning to improve understanding of ventilation and develop tools that improve patient outcomes, while collaborating with laboratory scientists to understand the biological basis for our findings. The field of data science in critical care is expanding but limited work has been performed in ventilation. This Fellowship will enable the curation of one of the largest longitudinal datasets for critical care globally, enabling future analyses and creating an open dataset for other investigators and collaborators.

Securing this NIHR Fellowship will provide the necessary protected time to immerse myself in this research. The Fellowship will offer accelerated personal and career development to develop research skills, while providing access to the NIHR infrastructure, such as career development schemes, mentorship, and networking opportunities, aiding development as an independent investigator. The Fellowship will allow me to work within a world-leading critical care research group at Imperial, while accessing collaboration opportunities. The support and endorsement of the NIHR will be invaluable during dissemination of our findings to the international research community, clinicians, patients, and the public, which will enhance my clinical, data science, and academic networks. All these experiences will strengthen my academic portfolio and help with the transition to working as an independent researcher.

RESEARCH SUPPORT / SUPERVISOR 1

Name	Dr Brijesh Patel

Institution	Imperial College London

Position	Consultant in Anaesthesia, Critical Care and Extracorporeal Life Support, Clinical Senior Lecturer

Description of support provided
Dr Patel was my supervisor during my NIHR Academic Clinical Fellowship, and will be my primary supervisor during the PhD, maintaining an overview of the project. He has a background in basic and translational research in respiratory failure and mechanical ventilation, with an interest in understanding the biological and physiological mechanisms of failure to wean from the ventilator. He has been awarded > £5 million in translational critical care grants, and is recognised as an international expert in the field, with > 80 publications and several national and international awards. He led the national service evaluation during the COVID-19 pandemic, which used explainable artificial intelligence to improve understanding of this disease. He has supervised 7 PhD candidates, of which 4 have been awarded a PhD, 1 in write up phase, and he is currently supervising 2 candidates. He has also led two clinical trials (in collaboration with Professor Rees), on the implementation of clinical decision support systems in critical care, which have led to the development of this proposal. Furthermore, he is principal investigator for a NIHR EME funded RCT examining mechanical ventilation strategies in patients with severe respiratory failure supported on extracorporeal life support, and studies examining ICU-acquired muscle wasting, a critical determinant and consequence of failure to wean. I will meet with Dr Patel on a weekly basis, and he will provide support on all aspects of this project, maintaining high-level oversight.

RESEARCH SUPPORT / SUPERVISOR 2

Name	Professor Padmanabhan Ramnarayan

Institution	Imperial College London

Position	Professor in Paediatric Critical Care, Honorary Consultant in Paediatric Critical Care

Description of support provided
Dr Ramnarayan's research interests are focused on the use of artificial intelligence to improve decision-making in critically ill children, use of novel non-invasive imaging techniques such as lung ultrasound during ventilation, and pragmatic clinical trials, particularly related to paediatric ventilation. He led a master protocol of two national clinical trials on non-invasive ventilation in children (FIRST-ABC) and he is currently leading a clinical trial of respiratory support in acute bronchiolitis (BACHb). He has an active interest in clinical informatics, which started during his MD(Res) research conducting a series of studies examining the accuracy and utility of a computerised diagnostic decision support system. He has been awarded £4 million in critical care research grants as Chief Investigator and ~£14 million as Co-Investigator, and he is recognised as an international expert in the field, with over 150 publications. He is currently supervising 4 PhD candidates and 1 MD(Res) candidate, one of whom is due for completion in 2024. Dr Ramnarayan will provide support on the paediatric ventilation aspects of this project, which has traditionally been considered separately to adult ventilation, as well as lending his expertise on decision support systems.

RESEARCH SUPPORT / SUPERVISOR 3

Name	Dr Melody Ni

Institution	Imperial College London

Position	Senior Research Fellow

Description of support provided
Dr Ni is a decision analyst and a methodologist with over 17 years of experiences in generating multi-dimensional evidence for the translation of innovative ideas into clinical practice. Dr Ni is a deputy director of the NIHR HRC in IVD at Imperial and has been leading a team of multi-disciplinary researchers since 2013, delivering impact through a range of innovative projects, including the development of a breath test for the detection of gastrointestinal cancers. Dr Ni chairs a regulatory network for in vitro diagnostics, working with industry partners and key stakeholders including MHRA and NICE. She has also hosted multiple accelerator programmes aimed at building knowledge and expertise for clinical scientists and innovators. Dr Ni has won over £10 million in grant funding, recognised as an international expert in the field, with over 60 peer-reviewed publications. She has supervised 7 PhD students, including 2 as lead supervisor, of which 6 have been awarded a PhD and 1 an MD. She is currently supervising 3 PhD candidates, 1 as lead supervisor. Dr Ni will provide expertise and guidance on the methods and approach of the project, both quantitatively and qualitatively, focusing on the real world aspect of the technology and the value a solution can realistically deliver. She also provides links to NIHR centres including the HRCs.

RESEARCH SUPPORT / SUPERVISOR 4

Name	Dr Ovidiu Serban

Institution	Imperial College London

Position	Research Fellow

Description of support provided
Dr Serban is a Research Fellow at the Imperial Data Science Institute and the Data Observatory group lead. His research interests include machine learning, natural language processing, affective computing and interactive

system design. His work includes real-time natural language processing, data curation and large-scale visualisation systems. He holds a joint PhD from INSA de Rouen Normandy (France) and "Babes-Bolyai" University in Romania while working at the LITIS Laboratory in France. He has over 40 peer-reviewed publications, including one best paper award at a NeurIPS workshop. He has supervised 3 PhD students as lead supervisor, of which two are close to their viva, and one as a co-supervisor. He supervised over 40 MSc students at Imperial College, with over 70% receiving distinction or distinguished, including three students being awarded president or department awards. Dr Serban will oversee the project's machine learning and data visualisation aspects and provide a link to the Data Observatory at the DSI.

Collaborations

I have built several important collaborations. I will partner with the Imperial Data Science Institute (DSI) on the data science and visualisation aspects. Their aim is to advance research and create a data science community; the unifying theme of their research is innovation for building, exploring, and exploiting modern datasets. They host researchers from multiple disciplines, including Biomedical Informatics, Data Visualisation and AI System Architecture. Specifically, I will be supervised by Dr Ovidiu Serban, an expert in machine learning and large-scale visualisation, and work with Dr Kai Sun, who has experience managing translational research projects. They will provide in-kind access to the Data Observatory throughout the project for visualisation testing. This collaboration supports two of the core objectives of the DSI (developing data analysis technology that supports research and translating data science into innovation through public sector partnership); it also supports the DSI's portfolio expansion in healthcare data, while leading to new relationships with academics from Imperial College London.

I will collaborate with the NIHR Healthtech Research Centre (HRC), whose aim is to support the development of medical devices, diagnostics, and digital technologies, which help clinicians and patients to monitor, diagnose and manage conditions including cancer and cardiovascular and respiratory diseases. Their multidisciplinary team includes experts in clinical, technology, human factors and health economics, who drive innovation, advance new technology and support healthcare staff to reduce workload. I will be supervised by Dr Melody Ni, deputy director of the NIHR-HRC. The HRC will also assist with building networks with world-leading hospitals and clinical academics. My project will add to their portfolio of innovation using artificial intelligence and the development of decision-aiding medical devices.

I will mature an exciting collaboration with Aalborg University, which was established during an introductory visit in August 2022. I worked with Professor Steve Rees, Professor of Respiratory Critical Care Technology in the Department of Health Science and Technology, on a preliminary analysis using physiological data to subgroup patients weaning from the ventilator in the Weaning Algorithms for mechanical VEntilation (WAVE, NCT 03697785) study. Subsequently, we have worked on a narrative review assessing clinical practice and decision-making in patients on the ventilator (published in the *British Journal of Anaesthesia*). Professor Rees and his research team have led the development of a model-based clinical decision support system for mechanical ventilation, the Beacon Caresystem, which has been shown to provide appropriate ventilator settings, reduce pressure support whilst preserving respiratory function, and been the subject of several international randomised controlled trials, during which they have collaborated with clinicians at Imperial College London, including my primary supervisor, Dr Brijesh Patel. I intend to mature this collaboration during the PhD with a second visit to Aalborg, which has been costed into this proposal, to develop expertise in data analysis techniques and decision support; the Aalborg team provide expertise in biomedical engineering, physiological modelling, and decision support systems. Aalborg University foresee several advantages to the collaboration, including improved understanding of respiratory physiology, exploring novel methods for improving explainability of decision support systems, while fostering ongoing international networks.

Finally, I have several partnerships with NHS and university collaborators, including with Dr TC Aw at the Royal Brompton Hospital, an expert in real-world clinical data development, Professor Luigi Camporota, a clinical and academic expert in mechanical ventilation, Dr Tim Stephens, Clinical Lecture and Implementation Scientist at Queen Mary University of London, an expert in qualitative research and methodology, and Dr Sonali Parbhoo, Assistant Professor and leader of the AI for Actionable Impact lab at Imperial, an expert in machine learning. Their guidance and support will be key to ensuring the success of this Fellowship.

Host Organisation Support Statement

The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support

I am writing as Head of the Department of Surgery & Cancer, Imperial College London (ICL), to express my full support Dr Mayur Murali in his application for an NIHR Doctoral Fellowship, entitled '*BREATHE: **B**etter **R**espiratory w**EA**ning using **T**ransparent mac**H**ine learning and data visualisation for invasiv**E** mechanical ventilation*'.

Mayur recently completed a NIHR Academic Clinical Fellowship (ACF) at ICL within the Division of Anaesthesia, Pain Medicine, and Intensive Care (APMIC). Over three years he developed research skills in mechanical ventilation and data science, conducting work forming the basis for his PhD application. This includes a narrative review (published in the *British Journal of Anaesthesia*), a national survey and preliminary analysis of ventilation data from a randomised controlled trial of 200+ patients. He also developed a long-term collaboration with the Department of Health, Science and Technology at Aalborg University, focused on ventilatory clinical decision support. He has further integrated with core capability at ICL including developing new areas of investigation with the Imperial-NIHR Healthtech Research Centre (Imperial-NIHR HRC), and the Imperial Data Science Institute (DSI). Additionally, Dr Murali has completed a Master's in Health Data Science at University College London (UCL) with merit, learning to code in R, Python, SQL, and Stata, and developing skills in statistics, analysis of large datasets, and machine learning. He is a national coordinator for GEMINI, an international study assessing ventilation practices involving approximately 10,000 patients, with the first abstract co-authored by Dr Murali, scheduled for presentation at the 2025 American Thoracic Society International Conference.

He recently completed a Trainee Editorial Fellowship with *Anaesthesia*, editing over 40 manuscripts, contributing to the Editorial Board and presenting at conferences. This complements a strong academic record, including 12 publications, 12 abstracts, co-editing and publishing a book, active participation in trainee research networks, including as Chair of the Pan-London Perioperative Audit & Research Network earning collaboration in six publications. He has developed teaching programmes, including the development of online modules for the British Medical Journal and Association of Anaesthetists, and earned a Master's in Medical Education. Mayur is also a Fellow of the South Asian Health Foundation, where he has worked on health inequality projects, deepening his commitment to equality, diversity and inclusion (EDI) in research. These efforts led to recognition through multiple awards, including a Higher Education Academy Fellowship, alongside clinical training and postgraduate exams. Additionally, Mayur has gained valuable insight into healthcare policy and health services research during a leadership fellowship with NHS England. The skills he has learned through these numerous and varied achievements will enable him to successfully deliver this proposal.

The proposed project will investigate whether data visualisation of multi-horizon forecasting can effectively characterise subgroups of patients undergoing ventilator weaning. By analysing longitudinal data, the goal is to develop an artificial intelligence (AI)-based tool to support bedside decision-making. This innovative approach seeks to develop explainable AI, while simultaneously enhancing the scientific understanding of weaning from mechanical ventilation – an essential part of critical care management, where diverse practice exists. This project builds upon Mayur's prior work during his ACF and has the potential to be applied to other phases of ventilation and translate to the development of a clinical decision support system for mechanical ventilation. Mayur has actively engaged patient and public partners throughout the project design, consulting multiple groups with experience in both adult and paediatric critical care. His application is further strengthened by the support of a PPI co-applicant with lived experience of intensive care.

The Department has a strong track record of nurturing students and trainees into independent scientists. Its tailored support and mentorship infrastructure will enable Mayur to maximise his potential. His lead supervisor, Dr Brijesh Patel, is an international expert in in respiratory failure and mechanical ventilation, with an interest in understanding mechanisms of failure to wean from the ventilator. He is director of the critical care and severe respiratory failure and ventilation service at the Royal Brompton Hospital and was lead for the national service COVID-19 evaluations using machine learning. His co-supervisors include Dr Padmanabhan Ramnarayan, Professor in Paediatric Critical Care, expert in paediatric ventilation and AI in critically ill children; Dr Melody Ni, senior methodologist within the Imperial-NIHR HRC and leader in health technology translation; and Dr Ovidiu Serban, Research Fellow at the Imperial DSI and lead of the Data Observatory Group, expert in machine learning, data curation and largescale visualisation systems. Collaborators from the University of Aalborg, and NHS partners (ICHT, GSTT, GOSH, Barts Health) will further in support Mayur's development. Initiatives within the Clinical Academic Training Office (CATO) including mentorship, networking and training resources will enhance his career development.

This proposal, including adults and children, supports the departmental strategy to bring together adult and paediatric critical care researchers under one umbrella to expand the impact of our research. The wider critical

care research team, of which Mayur is now a part, is recognised as a front-runner in data science and critical care research, including developing the AI Clinician for the management of patients with sepsis. Mayur has also developed collaborations which provide the necessary expertise in data science and visualisation, implementation science and engineering. The Department of Surgery & Cancer is keen to strengthen its AI research portfolio and as such, gives its full support to this research project.

ICL and the Department of Surgery & Cancer, have a diverse community of staff and students where everyone is treated with respect and given opportunities to flourish, both personally and professionally. This commitment to EDI is reflected in the Department's recent Athena Swan Gold Award in 2024, and broader EDI accreditations, including the Race Equality Charter, Time to Change, Disability Confident, AccessAble, and Stonewall. This open, transparent research environment will help Mayur grow as a researcher and leader.

I strongly believe that Mayur will execute his research proposal to a high standard, laying a foundation for future research and career growth. He possesses the attributes to maximise the benefits of an NIHR Fellowship and I fully support his application.

Your Sincerely,

Professor George Hanna

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Mayur Murali

Employing organisation	Imperial College of Science, Technology and Medicine

Grade	ST7

Starting point on scale	5

Increment Date	

Justification of Costs
Full-time salary for research post, lasting 3 years. This salary has been calculated according to my grade as an ST7 anaesthesia and intensive care registrar in a training programme, with the London Allowance of £2,162. https://www.bma.org.uk/pay-and-contracts/pay/resident-doctors-pay-scales/pay-scales-for-resident-doctors-in-england

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	70425
Geographical Weighting (£)	2162
Other allowances (£)	362.94
Superannuation (£)	11744.58
Nat. Insurance (£)	10138.05
Current Annual Costs (£)	**94832.57**

2

Staff type	Shared Staff

Role on this research	Research Engineer

Name	To be appointed

Employing organisation	Imperial College of Science, Technology and Medicine

Grade	Level 3a

Starting point on scale	22

Increment Date	

Justification of Costs
Will provide technical support for the development of data visualisations at the Data Science Institute Data Observatory during the Fellowship. The Data Observatory is the largest open-source facility of this nature in Europe; it is a tool that will allow the generation of new insights from complex datasets and provide a method of communicating our findings to clinicians and patients in an immersive environment.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	40201
Geographical Weighting (£)	
Other allowances (£)	201.01
Superannuation (£)	8442.21
Nat. Insurance (£)	5280.15
Current Annual Costs (£)	**54124.37**

Staff Type: (Lead Applicant)-Dr Mayur Murali
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 29.0
Justification of Cost: Full-time salary for research post, lasting 3 years. This salary has been calculated according to my grade as an ST7 anaesthesia and intensive care registrar in a training programme, with the London Allowance of £2,162. https://www.bma.org.uk/pay-and-contracts/pay/resident-doctors-pay-scales/pay-scales-for-resident-doctors-in-england

	Year 1	**Year 2**	**Year 3**	**Total Cost (£)**
Basic Salary	72587	72587	30245	175419
National Insurance	10138	10138	4224	24500
Superannuation	11745	11745	4894	28384
Geographical weighting / Other allowances (if applicable)	363	363	151	877

Staff Type: Other
Other Type: Shared Staff
Name: **Research Engineer**
Type of cost: HEI
% Full-time on this research: 4.0
Total months on this research: 29.0

Justification of Cost: Will provide technical support for the development of data visualisations at the Data Science Institute Data Observatory during the Fellowship. The Data Observatory is the largest open-source facility of this nature in Europe; it is a tool that will allow the generation of new insights from complex datasets and provide a method of communicating our findings to clinicians and patients in an immersive environment.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	1894	1934	823	4651

	Year 1	Year 2	Year 3	Total Cost (£)
Total HEI	77381	77413	32269	187063
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Collaborative work/ learning special techniques to Aalborg University
Justification of Cost: Travel to Aalborg University for 1 week to mature a collaboration with Professor Steve Rees, Professor of Respiratory Critical Care Technology, with expertise in CDSS and biomedical engineering. I will be developing skills in data analysis, machine learning, and decision support, enriching my development and that of our critical care research group in London, while fostering ongoing academic relationships. Travel including flights - £400, accommodation - £700, subsistence - £400. Total cost = £1,500

	Year 1	Year 2	Year 3	Total Cost (£)
	621	621	259	1501

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: 1 UK/ European per year/ 1 international conference: European Society of Intensive Care Medicine/ American Thoracic Society conference/ 1 UK conference
Justification of Cost: Conference fees, travel and subsistence to 3 conferences to present work during the Fellowship: - American Thoracic Society Conference (conference fee £550, travel + subsistence £1,000) - European Society of Intensive Care Medicine (conference fee approx £300, travel + subsistence £500) - UK Intensive Care Society State of the Art Conference (conference fee £450, travel + subsistence £200)

	Year 1	Year 2	Year 3	Total Cost (£)
	1242	1242	517	3001

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: PPIE - Conference attendance
Justification of Cost: Conference fees, travel, subsistence and payment for time for Patient Family & Advisory Group member (likely co-applicant, Ms Joanne Lloyd) to attend a conference to co-present our findings. - ICS SoA fee: £125 - £25/h for 8h at conference: £200 - Travel: £100 - Subsistence: £25 Total cost: £450

	Year 1	Year 2	Year 3	Total Cost (£)
	186	186	78	450

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	2049	2049	854	4952

Equipment

Type of cost: HEI
VAT relief to be applied: False
Description: Desktop computer with GPU
Price excluding VAT £: 1000
Expected life time: 36
Justification of Cost: Desktop computer with GPU for processing large datasets. Cost estimated from price on Imperial purchasing system with GPU specifications sufficient for data analysis (e.g. entry level Dell XPS desktop with appropriate CPU (12th Gen Intel i9), RAM (16 GB) and graphics card (GeForce RTX 3060 Ti 8GB) will cost at least £1000.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	414	414	172	1000

Type of cost: HEI
VAT relief to be applied: False
Description: Empatica EmbracePlus wristband and 3-year subscription
Price excluding VAT £: 1600
Expected life time: 36
Justification of Cost: Subscription for the Empatica EmbracePlus wristband, which will be used for work package 3, the simulation study. The wristband will measure physiological signals (e.g. electrodermal activity, skin temperature, heart rate variability) as an indicator of sympathetic stimulation and thus a proxy for cognitive load. Subscription includes access to the secure Cloud environment in which data is stored. Price obtained from website: https://www.empatica.com/platform/research-studies/

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	662	662	276	1600

Type of cost: HEI
VAT relief to be applied: False
Description: Google Cloud access
Price excluding VAT £: 840
Expected life time: 36
Justification of Cost: Access to secure Google Cloud platform where MIMIC IV and NHS databases are hosted. Having used the platform previously, average cost has been £10-20/ month, I expect this to increase as the scale of the project increases. Costed for £840 over 36 months of the project.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	348	348	145	841

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	1424	1424	593	3441

Summary

	Travel (excl Conferences)	Travel (Conferences)	Equipment
Total HEI	1201	3451	2753
Total NHS	0	0	0
Total Commercial	0	0	0
Total Other	0	0	0

Consumables

Description: Mobile phone, mobile tariff and Zoom account
Type of cost: HEI
Please describe fully: Mobile phone, mobile tariff and Zoom account
Justification of Cost: Mobile phone and tariff for study activities - costed via the Imperial purchasing system (cost of a simple Samsung handset (£600) and tariff for 36m (£10-15/m = approx £500)). Zoom subscription for PFAG meetings costed at £10.83/m according to website (https://zoom.us/pricing), approx £400 for 36m of Fellowship.

	Year 1	Year 2	Year 3	Total Cost (£)
	621	621	259	1501

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	621	621	259	1501

Patient and Public Involvement

Description: PFAG: 8 x 2h meetings over 28m/ PFAG: co-design workshops/ Co-applicant monthly meetings/Co-applicant to attend co-design workshops/ Co-applicant to assist with results dissemination/ Co-applicant to assist with co-design analysis
Type of cost: HEI
Please describe fully: PFAG: 8 x 2h meetings over 28m/ PFAG: co-design workshops/ Co-applicant monthly meetings/Co-applicant to attend co-design workshops/ Co-applicant to assist with results dissemination/ Co-applicant to assist with co-design analysis
Justification of Cost: Costed Patient Family & Advisory Group and co-applicant meetings as per the recommended rates from the NIHR payment guidance for researchers and professionals (https://www.nihr.ac.uk/payment-guidance-researchers-and-professionals). Within this, I have costed 4-monthly meetings for my 11-strong PFAG group, and 2 co-design workshops for 3 PFAG members. - 8 x 2h online meetings, including 1 hour preparation time and internet connection cost. PFAG members are paid at £25/h and internet access cost is £5. Therefore, cost is £80/person/meeting. Total cost = £80 x 11 x 8 = £7,040 - 2 x 4h in-person co-design workshops with no preparation time or internet cost attached, but travel cost included. PFAG members are paid at £25/h and travel is costed at £100/person. Therefore cost is £200/person/workshop. Total cost = £200 x 3 x 2 = £1,200 I have costed the following for my co-applicant: Meetings: - 20 x 1h online monthly meetings, costed at £25/h + 1 hour preparation time + internet access costs = £55/ meeting = £1,100 - 8 x 2h online meetings with PFAG, including 1 hour preparation time and internet connection cost. Co-applicant paid at £25/h and internet access cost is £5. Therefore, cost is £80/ meeting = £640 Total cost for meetings = £1,740 Co-design workshops: - 2 x 4h in-person co-design workshops with no preparation time or internet cost attached, but travel cost included. Co-applicant paid at £25/h and travel is costed at £100/workshop. Therefore cost is £200/person/workshop. Total cost = £400 Data analysis for co-design workshops: - 3 x full-day online meetings for analysis of co-design workshops, costed at £150/day (no preparation time required) Total cost = £150 x 3 = £450 Results dissemination: - 5 x full-day online meetings/ conference attendance to co-produce messaging for public dissemination/ attend conferences to co-present results, costed at £150/day (no preparation time required) Total cost = £150 x 5 = £750 TOTAL COST FOR PPIE/ CO-APPLICANT ACTIVITIES: £11,580

	Year 1	Year 2	Year 3	Total Cost (£)
	4792	4792	1996	11580

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	4792	4792	1996	11580

Other Direct Costs

Description: Dissemination of results (Digital)/ In person event for 100 people for dissemination of results
Type of cost: HEI
Please describe fully: Dissemination of results (Digital)/ In person event for 100 people for dissemination of results
Justification of Cost: Cost for results dissemination, including: Newsletter with results - £1,200 for design, £1,200 for online publication = £2,400 Social media - short video/ infographic summarising main findings - costed at £4,800 In-person event for 100 people for dissemination of results - room hire costed at £1,000, refreshments costed at £4/head = £1,400 Total cost = £8,600

	Year 1	Year 2	Year 3	Total Cost (£)
	3559	3559	1482	8600

Description: Research Nurse Guys & St Thomas NHS Foundation Trust
Type of cost: NHS
Please describe fully: Research Nurse Guys & St Thomas NHS Foundation Trust
Justification of Cost: Barnaby Sanderson, Research Nurse at Kings College Hospital, who has been costed for curating the Guy's and St Thomas' database. Costed at 2% for 12 months - Research Nurse Band 7 on the top step point

	Year 1	Year 2	Year 3	Total Cost (£)
	593	593	247	1433

Description: Dr from Royal Brompton & Harefield NHS Foundation Trust
Type of cost: NHS
Please describe fully: Dr from Royal Brompton & Harefield NHS Foundation Trust
Justification of Cost: Dr TC Aw, Associate Chief Clinical Information Officer at the Royal Brompton Hospital, an expert in real-world clinical data development, costed into the application for curation of the Royal Brompton Hospital database. Costed at 1% for 3 years - Consultant salary

	Year 1	Year 2	Year 3	Total Cost (£)
	1863	1863	776	4502

Description: Clinician time for WP2
Type of cost: HEI
Please describe fully: Clinician time for WP2
Justification of Cost: Work package 2 includes 1 focus group, 2 co-design workshops and 1 round of beta-testing. Clinicians have been costed as per the NIHR payment guide for researchers and professionals (https://www.nihr.ac.uk/payment-guidance-researchers-and-professionals), in keeping with payment for the PPIE members involved with the same activity, at £25/h. Clinicians for this work package will be drawn from London hospitals, so we have costed £25 of travel per person per workshop. Therefore: - 1 x 90m focus group: £25/h + £25 for travel for 20 clinicians to attend = £1,250 - 2 x 4h co-design workshops: £25/h + £25 for travel for 20 clinicians to attend = £5,000 - 1.x 60m of beta testing: £25/h + £25 travel for 10 clinicians to attend = £500 Total cost = £6,750

	Year 1	Year 2	Year 3	Total Cost (£)
	2793	2793	1163	6749

Description: Clinician time for WP3
Type of cost: HEI
Please describe fully: Clinician time for WP3
Justification of Cost: Work package 3 is a human-AI interaction vignette study using a modified, between-subjects design, with three arms, where each clinician will experience all three arms on different patient subsets. The three arms are: baseline with no additional AI information (baseline); additional AI CDSS information (AI); CDSS with XAI decision using data visualisation (XAI). The primary outcome will be difference in cognitive load between the three arms. A power calculation has shown that a sample of 102 clinicians will provide the experiment with a 90% power to detect a difference in mean physiological signals, assuming an effect size of 0.25 and a significance level of 0.05. Each experiment will last 1h, and we have costed each clinician's time at £50/h, based on previous work by our group, with £25 allocated for travel. Total cost = £75 x 102 clinicians = £7,650

	Year 1	Year 2	Year 3	Total Cost (£)
	3166	3166	1318	7650

Description: Facilitator for focus group, co-design workshops and beta testing
Type of cost: HEI
Please describe fully: Facilitator for focus group, co-design workshops and beta testing
Justification of Cost: Facilitation by Dr Tim Stephens, Clinical Lecture and Implementation Scientist at Queen Mary University of London, and an expert in qualitative research and methodology. Quoted £650/ focus group, co-design workshop or beta-testing event. Total focus groups: 3 Total co-design workshops: 6 Beta testing event: 1 Therefore, costed at £650 x 10 = £6,500

	Year 1	Year 2	Year 3	Total Cost (£)

	2690	2690	1120	6500

Description: Paediatric database plus 1 additional instance of data transfer
Type of cost: HEI
Please describe fully: Paediatric database plus 1 additional instance of data transfer
Justification of Cost: Access to the Big Data and Analytics Unit Secure Environment (BDAU SE), in which the aggregated paediatric database is hosted. Quoted £3,600 for access for 1 year, with an additional instance of data transfer costed at £300.

	Year 1	Year 2	Year 3	Total Cost (£)
	1614	1614	672	3900

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	16278	16278	6778	39334

Training And Development

Type of cost: HEI
Description: Short Courses
Item Description: Developing & Evaluating Complex Interventions - University of Glasgow/ Experience Based Co-design - Point of Care Foundation/ An Introduction to Implementation Science, KCL/ Qualitative Research Methods in Health, UCL/ Data Visualisation & Visual Analytics, Alan Turing Institute (free), Maths for ML: linear algebra, ICL
Justification of Cost: Courses: Developing & Evaluating Complex Interventions, University of Glasgow - £190 Experience Based Co-design, Point of Care Foundation - £395 An Introduction to Implementation Science, KCL - £1,875 Qualitative Research Methods in Health, UCL - £1,500 Data Visualisation & Visual Analytics, Alan Turing Institute - free Maths for ML: linear algebra, ICL (delivered via Coursera) - free Total cost = £3.960

	Year 1	Year 2	Year 3	Total Cost (£)
	1639	1639	682	3960

Type of cost: HEI
Description: Short Courses
Item Description: Online training courses: Datacamp/ Coursera
Justification of Cost: Datacamp subscription - £20/m for 36m = £720 Coursera - £304/ year = £912 Total cost = £1,632

	Year 1	Year 2	Year 3	Total Cost (£)
	675	675	281	1631

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: Tuition fees
Justification of Cost: Imperial College London tuition fees for undertaking a PhD are costed at £7,280/ year for a 3-year doctorate.

	Year 1	Year 2	Year 3	Total Cost (£)
	7280	7280	7280	21840

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	9594	9594	8243	27431

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	107883	107915	49220	265018

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: Imperial College of Science, Technology and Medicine
Justification of Cost: TBC

	Year 1	Year 2	Year 3	Total Cost (£)
	28098	28098	11707	67903

Type of cost: Other Indirect Costs
Name of Higher Education Institute: Imperial College of Science, Technology and Medicine
Justification of Cost: TBC

	Year 1	Year 2	Year 3	Total Cost (£)
	88087	88087	36703	212877

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	116185	116185	48410	280780

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	92948	92948	38728	224624

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	228201	224624	452825
Total Higher Education Institution Costs (100%)s	30882	0	30882
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	5935	0	5935
Total Research Costs Requested From Funder	265018	224624	489642

Summary

	Year 1	Year 2	Year 3	Total Cost (£)

Salaries	**77381**	**77413**	**32269**	**187063**
Travel-subsistence-and-conference-fees	**1925**	**1925**	**802**	**4652**
Equipment	**1139**	**1139**	**475**	**2753**
Consumables	**497**	**497**	**207**	**1201**
Patient-and-Public-Involvement	**3834**	**3834**	**1597**	**9265**
Other-Direct-Costs	**13513**	**13513**	**5627**	**32653**
Training And Development	**9594**	**9594**	**8243**	**27431**
HEI-Indirect-costs	**92948**	**92948**	**38728**	**224624**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**200831**	**200863**	**87948**	**489642**

NHS Support Costs
No costs have been added

Have you discussed and agreed these costs with the Lead Network?	N/A

NHS Treatment Costs
No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs
No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	**Year 1**	**Year 2**	**Year 3**	**Total Cost (£)**
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	489642
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	489642

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as a .csv file or Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - References.docx) is included as an appendix within this file.

(Research Timetable - Timeline_MM.pdf) is included as an appendix within this file.

(SoECAT Template - Funder Export_340376_SoECAT_BREATHE.xlsx) is included as an appendix within this file.

11. Participants and Signatory

Participants
New participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please see the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

You are required to supply the names and email addresses (if not already registered with this online application system) of each required participant. Everyone **newly** named in this section will be acting as a 'participant' to your application and will need to agree to be part of this application.

By confirming participation, participants are acknowledging their involvement and input into this application, and agreeing to be involved in the proposed fellowship. You must ensure all **new** participants are happy for your application to be submitted before submitting it on the online system.

It is permissible for a single individual to serve as a signatory and a participant, but they will need to be added separately under each heading.

Details of the individual who will be required to approve your application after submission should be entered in the 'Participants and Signatory' section. Please note that it is acceptable for a Head of Department to also act as a research support role. If this is the case, they will need to be included in this section for both roles.

Participants must confirm their participation on your application before you will be able to press the submit button. Please see the 'Application Submission Process Flow Diagram' in the guidance notes for further information.

Doctoral Primary Supervisor

Doctoral Primary Supervisor Participant

Title	Dr
First Name	Brijesh
Last Name	Patel
Position	Consultant / Clinical Senior Lecturer

Supervisor

Supervisor Participant

Title	Dr
First Name	Melody
Last Name	Ni
Position	Senior Research fellow

Supervisor Participant

Title	Professor
First Name	Padmanabhan
Last Name	Ramnarayan

Position	Professor in Paediatric Critical Care

Supervisor Participant

Title	Dr
First Name	Ovidiu
Last Name	Serban
Position	

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Ms
First Name	Jennifer
Last Name	Landmann
Position	Research Proposal Manager

Signatory
Participants must complete their actions on your application before submission; the signatory must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

You are required to supply the name and email address (if not already registered on the ARAMIS application system) of the individuals who will be 'signing off' your application. Once their contact details have been entered below, these individuals will be invited to log in and confirm their participation in your application.

The signatory must ALSO approve the application AFTER you have selected the 'SUBMIT' option but BEFORE the application submission deadline. Please see the Submission Process Flow Diagram and Applicant Guidance Notes for further information.

The signatory approval will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatory) will be notified of this via an automated system-generated email.
NIHR will not accept any applications unless fully approved by your signatory before the 1:00 pm deadline on Thursday 16th January 2025.

Host Organisation - Heads of Department

Head of Department or Senior Manager

Title	
First Name	Professor George
Last Name	Hanna
Position	Head of Division of Surgery

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
N/A

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR Research Governance Framework. If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. References.docx
2. Timeline_MM.pdf
3. Funder Export_340376_SoECAT_BREATHE.xlsx

Appendix #1: References.docx

References

1. Fan E, Brodie D, Slutsky AS. Acute Respiratory Distress Syndrome: Advances in Diagnosis and Treatment. *JAMA* 2018; 319: 698.

2. ICNARC. *Key statistics from the Case Mix Porgramme - adult, general critical care units*. Intensive Care National Audit & Research Centre, https://www.icnarc.org/Our-Audit/Audits/Cmp/Reports/Summary-Statistics (5 May 2022, accessed 19 November 2022).

3. PICANet. *PICANet State of the Nation Report 2023*, https://www.picanet.org.uk/annual-reporting-and-publications/ (2023).

4. Ventilation with Lower Tidal Volumes as Compared with Traditional Tidal Volumes for Acute Lung Injury and the Acute Respiratory Distress Syndrome. *N Engl J Med* 2000; 342: 1301–1308.

5. Guérin C, Reignier J, Richard J-C, et al. Prone Positioning in Severe Acute Respiratory Distress Syndrome. *N Engl J Med* 2013; 368: 2159–2168.

6. The RECOVERY Collaborative Group. Dexamethasone in Hospitalized Patients with Covid-19. *N Engl J Med* 2021; 384: 693–704.

7. on behalf of the United Kingdom COVID-ICU National Service Evaluation, Patel BV, Haar S, et al. Natural history, trajectory, and management of mechanically ventilated COVID-19 patients in the United Kingdom. *Intensive Care Med* 2021; 47: 549–565.

8. Morris AH, Stagg B, Lanspa M, et al. Enabling a learning healthcare system with automated computer protocols that produce replicable and personalized clinician actions. *J Am Med Inform Assoc* 2021; 28: 1330–1344.

9. Karbing DS, Spadaro S, Dey N, et al. An Open-Loop, Physiologic Model–Based Decision Support System Can Provide Appropriate Ventilator Settings: *Crit Care Med* 2018; 46: e642–e648.

10. Sutton RT, Pincock D, Baumgart DC, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. *Npj Digit Med* 2020; 3: 17.

11. Clavieras N, Wysocki M, Coisel Y, et al. Prospective Randomized Crossover Study of a New Closed-loop Control System *versus* Pressure Support during Weaning from Mechanical Ventilation. *Anesthesiology* 2013; 119: 631–641.

12. Lellouche F, Mancebo J, Jolliet P, et al. A Multicenter Randomized Trial of Computer-driven Protocolized Weaning from Mechanical Ventilation. *Am J Respir Crit Care Med* 2006; 174: 894–900.

13. East TD, Heermann LK, Bradshaw RL, et al. Efficacy of computerized decision support for mechanical ventilation: results of a prospective multi-center randomized trial. *Proc AMIA Symp* 1999; 251–255.

14. Su J, Lin C-Y, Chen P-J, et al. Experience with a step-down respiratory care center at a tertiary referral medical center in Taiwan. *J Crit Care* 2006; 21: 156–161.

15. Hsu J-C, Chen Y-F, Chung W-S, et al. Clinical Verification of A Clinical Decision Support System for Ventilator Weaning. *Biomed Eng OnLine* 2013; 12: S4.

16. Zheng K, Kumar S, Sarti AJ, et al. Economic feasibility of a novel tool to assist extubation decision-making: an early health economic modeling. *Int J Technol Assess Health Care* 2022; 38: e66.

17. Capdevila X, Perrigault P-F, Ramonatxo M, et al. Changes in breathing pattern and respiratory muscle performance parameters during difficult weaning: *Crit Care Med* 1998; 26: 79–87.

18. Vassilakopoulos T, Zakynthinos S, Roussos C. The Tension–Time Index and the Frequency/ Tidal Volume Ratio Are the Major Pathophysiologic Determinants of Weaning Failure and Success. *Am J Respir Crit Care Med* 1998; 158: 378–385.

19. Patel B, Mumby S, Johnson N, et al. *Decision support system to evaluate VENTilation in the Acute Respiratory Distress Syndrome*. Preprint, Intensive Care and Critical Care Medicine. Epub ahead of print 29 September 2023. DOI: 10.1101/2023.09.28.23295668.

20. Morris AH. Developing and Implementing Computerized Protocols for Standardization of Clinical Decisions. *Ann Intern Med* 2000; 132: 373.

21. Knighton AJ, Kean J, Wolfe D, et al. Multi-factorial barriers and facilitators to high adherence to lung-protective ventilation using a computerized protocol: a mixed methods study. *Implement Sci Commun* 2020; 1: 67.

22. Sarti AJ, Zheng K, Herry CL, et al. Feasibility of implementing *Extubation Advisor* , a clinical decision support tool to improve extubation decision-making in the ICU: a mixed-methods observational study. *BMJ Open* 2021; 11: e045674.

23. Liao K-M, Ko S-C, Liu C-F, et al. Development of an Interactive AI System for the Optimal Timing Prediction of Successful Weaning from Mechanical Ventilation for Patients in Respiratory Care Centers. *Diagnostics* 2022; 12: 975.

24. Lin M-Y, Li C-C, Lin P-H, et al. Explainable Machine Learning to Predict Successful Weaning Among Patients Requiring Prolonged Mechanical Ventilation: A Retrospective Cohort Study in Central Taiwan. *Front Med* 2021; 8: 663739.

25. Otaguro T, Tanaka H, Igarashi Y, et al. Machine Learning for Prediction of Successful Extubation of Mechanical Ventilated Patients in an Intensive Care Unit: A Retrospective Observational Study. *J Nippon Med Sch* 2021; 88: 408–417.

26. Fernando S, Amador Díaz López J, Şerban O, et al. Towards a large-scale twitter observatory for political events. *Future Gener Comput Syst* 2020; 110: 976–983.

27. McGinn D, Birch D, Akroyd D, et al. Visualizing Dynamic Bitcoin Transaction Patterns. *Big Data* 2016; 4: 109–119.

28. Molina-Solana M, Birch D, Guo Y. Improving data exploration in graphs with fuzzy logic and large-scale visualisation. *Appl Soft Comput* 2017; 53: 227–235.

29. Șerban O, Thapen N, Maginnis B, et al. Real-time processing of social media with SENTINEL: A syndromic surveillance system incorporating deep learning for health classification. *Inf Process Manag* 2019; 56: 1166–1184.

30. Islam H, Li X, Zaman Rana MdK, et al. An In-depth Needs Analysis to Design a Data Visualization Dashboard Prototype for Critical Care Setting. In: *2022 Workshop on Visual Analytics in Healthcare (VAHC)*. Washington, DC, USA: IEEE, pp. 01–06.

31. Brich N, Schulz C, Peter J, et al. Visual Analytics of Multivariate Intensive Care Time Series Data. *Comput Graph Forum* 2022; 41: 273–286.

32. Thomas MM, Kannampallil T, Abraham J, et al. Echo: A large display interactive visualization of ICU data for effective care handoffs. In: *2017 IEEE Workshop on Visual Analytics in Healthcare (VAHC)*. Phoenix, AZ: IEEE, pp. 47–54.

33. Key statistics from the Case Mix Programme - adult, general critical care units. 1st January 2019-31st December 2022., https://www.icnarc.org/Our-Audit/Audits/Cmp/Reports/Summary-Statistics (2022, accessed 7 July 2023).

34. Goligher E, Ferguson ND. Mechanical ventilation: epidemiological insights into current practices: *Curr Opin Crit Care* 2009; 15: 44–51.

35. Wunsch H, Linde-Zwirble WT, Angus DC, et al. The epidemiology of mechanical ventilation use in the United States*: *Crit Care Med* 2010; 38: 1947–1953.

36. Gunasekera P, Gratrix A. Ventilator-associated pneumonia. *BJA Educ* 2016; 16: 198–202.

37. Jubran A, Grant BJB, Duffner LA, et al. Long-Term Outcome after Prolonged Mechanical Ventilation. A Long-Term Acute-Care Hospital Study. *Am J Respir Crit Care Med* 2019; 199: 1508–1516.

38. NHS England. NHS Long Term Workforce Plan, https://www.england.nhs.uk/publication/nhs-long-term-workforce-plan/ (2023, accessed 3 December 2023).

39. The NHS Long Term Plan, https://www.longtermplan.nhs.uk/ (2019, accessed 7 July 2023).

40. Health Education England. The Topol Review, https://topol.hee.nhs.uk/ (2019, accessed 3 December 2023).

41. Tume LN, Kneyber MCJ, Blackwood B, et al. Mechanical Ventilation, Weaning Practices, and Decision Making in European PICUs*: *Pediatr Crit Care Med* 2017; 18: e182–e188.

42. Blackwood B, Junk C, Lyons JDM, et al. Role Responsibilities in Mechanical Ventilation and Weaning in Pediatric Intensive Care Units: A National Survey. *Am J Crit Care* 2013; 22: 189–197.

43. Rose L, Blackwood B, Egerod I, et al. Decisional responsibility for mechanical ventilation and weaning: an international survey. *Crit Care* 2011; 15: R295.

44. Alkhathami M, Al Haddad M, Alenazi M. PERCEIVED RESPONSIBILITY FOR MECHANICAL VENTILATION AND WEANING DECISIONS IN INTENSIVE CARE UNITS IN THE KINGDOM OF SAUDI ARABIA. *Chest* 2022; 161: A497.

45. Villa G, Manara D, Palese A. Nurses' near-decision-making process of postoperative patients' cardiosurgical weaning and extubation in an Italian environment. *Intensive Crit Care Nurs* 2012; 28: 41–49.

46. Rose L, Nelson S, Johnston L, et al. Workforce profile, organisation structure and role responsibility for ventilation and weaning practices in Australia and New Zealand intensive care units. *J Clin Nurs* 2008; 17: 1035–1043.

47. Curtis B, Rak K, Richardson A, et al. Understanding Interprofessional Decision-Making Processes and Perceptions of Oxygenation for Acute Respiratory Failure. In: *D106. STAFFING, TRAINING, AND IMPLEMENTATION TO IMPROVE ICU QUALITY*. American Thoracic Society, pp. A7814–A7814.

48. Haugdahl HS, Storli S, Rose L, et al. Perceived decisional responsibility for mechanical ventilation and weaning: a Norwegian survey: A Norwegian survey. *Nurs Crit Care* 2014; 19: 18–25.

49. Taylor F. A comparative study examining the decision-making processes of medical and nursing staff in weaning patients from mechanical ventilation. *Intensive Crit Care Nurs* 2006; 22: 253–263.

50. Balas MC, Tate J, Tan A, et al. Evaluation of the Perceived Barriers and Facilitators to Timely Extubation of Critically Ill Adults: An Interprofessional Survey. *Worldviews Evid Based Nurs* 2021; 18: 201–209.

51. Arnal J-M, Garnero A, Novotni D, et al. Closed loop ventilation mode in Intensive Care Unit: a randomized controlled clinical trial comparing the numbers of manual ventilator setting changes. *Minerva Anestesiol*; 84. Epub ahead of print January 2018. DOI: 10.23736/S0375-9393.17.11963-2.

52. Kydonaki K, Huby G, Tocher J, et al. Understanding nurses' decision-making when managing weaning from mechanical ventilation: a study of novice and experienced critical care nurses in Scotland and Greece. *J Clin Nurs* 2016; 25: 434–444.

53. Tingsvik C, Johansson K, Mårtensson J. Weaning from mechanical ventilation: factors that influence intensive care nurses' decision-making: Intensive care nurses' decision-making when weaning from mechanical ventilation. *Nurs Crit Care* 2015; 20: 16–24.

54. Lavelle C, Dowling M. The factors which influence nurses when weaning patients from mechanical ventilation: Findings from a qualitative study. *Intensive Crit Care Nurs* 2011; 27: 244–252.

55. Khalafi A, Elahi N, Ahmadi F. Continuous care and patients' basic needs during weaning from mechanical ventilation: A qualitative study. *Intensive Crit Care Nurs* 2016; 37: 37–45.

56. Rose L, Dainty KN, Jordan J, et al. Weaning From Mechanical Ventilation: A Scoping Review of Qualitative Studies. *Am J Crit Care* 2014; 23: e54–e70.

57. Hancock HC, Easen PR. The decision-making processes of nurses when extubating patients following cardiac surgery: An ethnographic study. *Int J Nurs Stud* 2006; 43: 693–705.

58. Gelsthorpe T, Crocker C. A study exploring factors which influence the decision to commence nurse-led weaning. *Nurs Crit Care* 2004; 9: 213–221.

59. Cederwall C-J, Plos K, Rose L, et al. Critical care nurses management of prolonged weaning: an interview study: Prolonged weaning. *Nurs Crit Care* 2014; 19: 236–242.

60. Mårtensson IE, Fridlund B. Factors influencing the patient during weaning from mechanical ventilation: a national survey. *Intensive Crit Care Nurs* 2002; 18: 219–229.

61. Pettersson S, Melaniuk-Bose M, Edell-Gustafsson U. Anaesthetists' perceptions of facilitative weaning strategies from mechanical ventilator in the intensive care unit (ICU): A qualitative interview study. *Intensive Crit Care Nurs* 2012; 28: 168–175.

62. Blackwood B, Wilson-Barnett J, Trinder J. Protocolized weaning from mechanical ventilation: ICU physicians' views. *J Adv Nurs* 2004; 48: 26–34.

63. Tulaimat A, Mokhlesi B. Accuracy and Reliability of Extubation Decisions by Intensivists. *Respir Care* 2011; 56: 920–927.

64. Allerød C, Karbing DS, Thorgaard P, et al. Variability of preference toward mechanical ventilator settings: A model-based behavioral analysis. *J Crit Care* 2011; 26: 637.e5-637.e12.

65. Aboodi MS, Chen J-T, Hope AA, et al. Exploring the Clinical Decision-Making Process in Approaches to Fluid Resuscitation and Vasopressor Administration in Septic Shock: A Qualitative Think-Aloud Study. In: *D25. CRITICAL CARE: HARD TIMES - RESUSCITATING MY PATIENT: FLUID, BLOOD AND OTHER STRATEGIES*. American Thoracic Society, pp. A6004–A6004.

66. Hakim R, Watanabe-Tejada L, Sukhal S, et al. Acute respiratory failure in randomized trials of noninvasive respiratory support: A systematic review of definitions, patient characteristics, and criteria for intubation. *J Crit Care* 2020; 57: 141–147.

67. Moraes F da S, Marengo LL, Moura MDG, et al. ABCDE and ABCDEF care bundles: A systematic review of the implementation process in intensive care units. *Medicine (Baltimore)* 2022; 101: e29499.

68. Nurses of the Central and General ICUs of Shiraz Namazi Hospital, Sedation an Delirium Group Hospital Universitari de Bellvitge, SPACeR group (Surrey Peri-operative, Anaesthesia and Critical Care Collaborative Research Group), et al. ESICM LIVES 2016: part two: Milan, Italy. 1–5 October 2016. *Intensive Care Med Exp* 2016; 4: 30, s40635-016-0099–9.

69. Calfee CS, Delucchi K, Parsons PE, et al. Subphenotypes in acute respiratory distress syndrome: latent class analysis of data from two randomised controlled trials. *Lancet Respir Med* 2014; 2: 611–620.

70. Calfee CS, Delucchi KL, Sinha P, et al. Acute respiratory distress syndrome subphenotypes and differential response to simvastatin: secondary analysis of a randomised controlled trial. *Lancet Respir Med* 2018; 6: 691–698.

71. Famous KR, Delucchi K, Ware LB, et al. Acute Respiratory Distress Syndrome Subphenotypes Respond Differently to Randomized Fluid Management Strategy. *Am J Respir Crit Care Med* 2017; 195: 331–338.

72. Writing Group for the Alveolar Recruitment for Acute Respiratory Distress Syndrome Trial (ART) Investigators, Cavalcanti AB, Suzumura ÉA, et al. Effect of Lung Recruitment and Titrated Positive End-Expiratory Pressure (PEEP) vs Low PEEP on Mortality in Patients With Acute Respiratory Distress Syndrome: A Randomized Clinical Trial. *JAMA* 2017; 318: 1335.

73. Bellani G, Guerra L, Musch G, et al. Lung Regional Metabolic Activity and Gas Volume Changes Induced by Tidal Ventilation in Patients with Acute Lung Injury. *Am J Respir Crit Care Med* 2011; 183: 1193–1199.

74. Gattinoni L, Caironi P, Cressoni M, et al. Lung Recruitment in Patients with the Acute Respiratory Distress Syndrome. *N Engl J Med* 2006; 354: 1775–1786.

75. Briel M, Meade M, Mercat A, et al. Higher vs Lower Positive End-Expiratory Pressure in Patients With Acute Lung Injury and Acute Respiratory Distress Syndrome: Systematic Review and Meta-analysis. *JAMA* 2010; 303: 865.

76. Constantin J-M, Jabaudon M, Lefrant J-Y, et al. Personalised mechanical ventilation tailored to lung morphology versus low positive end-expiratory pressure for patients with acute respiratory distress syndrome in France (the LIVE study): a multicentre, single-blind, randomised controlled trial. *Lancet Respir Med* 2019; 7: 870–880.

77. Bos LDJ, Sjoding M, Sinha P, et al. Longitudinal respiratory subphenotypes in patients with COVID-19-related acute respiratory distress syndrome: results from three observational cohorts. *Lancet Respir Med* 2021; 9: 1377–1386.

78. Toal CM, Fowler AJ, Patel BV, et al. Hypoxemia Trajectory of Non-COVID-19 Acute Respiratory Distress Syndrome Patients. An Observational Study Focusing on Hypoxemia Resolver Status. *Crit Care Explor* 2023; 5: e0985.

79. Tams CG, Ataya A, Euliano NR, et al. Decision support system facilitates rapid decreases in pressure support and appropriate inspiratory muscle workloads in adults with respiratory failure. *J Crit Care* 2017; 42: 213–217.

80. Banner MJ, Euliano NR, MacIntyre NR, et al. Ventilator Advisory System Employing Load and Tolerance Strategy Recommends Appropriate Pressure Support Ventilation Settings. *Chest* 2008; 133: 697–703.

81. Blackwood B, Ringrow S, Clarke M, et al. A Core Outcome Set for Critical Care Ventilation Trials. *Crit Care Med* 2019; 47: 1324–1331.

82. Saunders CH, Sierpe A, Plessen C von, et al. Practical thematic analysis: a guide for multidisciplinary health services research teams engaging in qualitative analysis. *BMJ* 2023; 381: e074256.

83. Johnson MP, Ballie J, Thorup T, et al. CO/DEsign: building a shared dialogue around analysis within co-design. *Des J* 2017; 20: S4241–S4252.

84. Montori VM, Breslin M, Maleska M, et al. Creating a Conversation: Insights from the Development of a Decision Aid. *PLOS Med* 2007; 4: e233.

85. Coulter A, Stilwell D, Kryworuchko J, et al. A systematic development process for patient decision aids. *BMC Med Inform Decis Mak* 2013; 13 Suppl 2: S2.

86. Weiner BJ, Lewis CC, Stanick C, et al. Psychometric assessment of three newly developed implementation outcome measures. *Implement Sci* 2017; 12: 108.

87. Wherton J, Sugarhood P, Procter R, et al. Co-production in practice: how people with assisted living needs can help design and evolve technologies and services. *Implement Sci* 2015; 10: 75.

88. EmbracePlus | The world’s most advanced smartwatch for continuous health monitoring. *Empatica*, https://www.empatica.com/embraceplus (accessed 2 January 2025).

89. Blackmore KL, Smith SP, Bailey JD, et al. Integrating Biofeedback and Artificial Intelligence into eXtended Reality Training Scenarios: A Systematic Literature Review. *Simul Gaming* 2024; 55: 445–478.

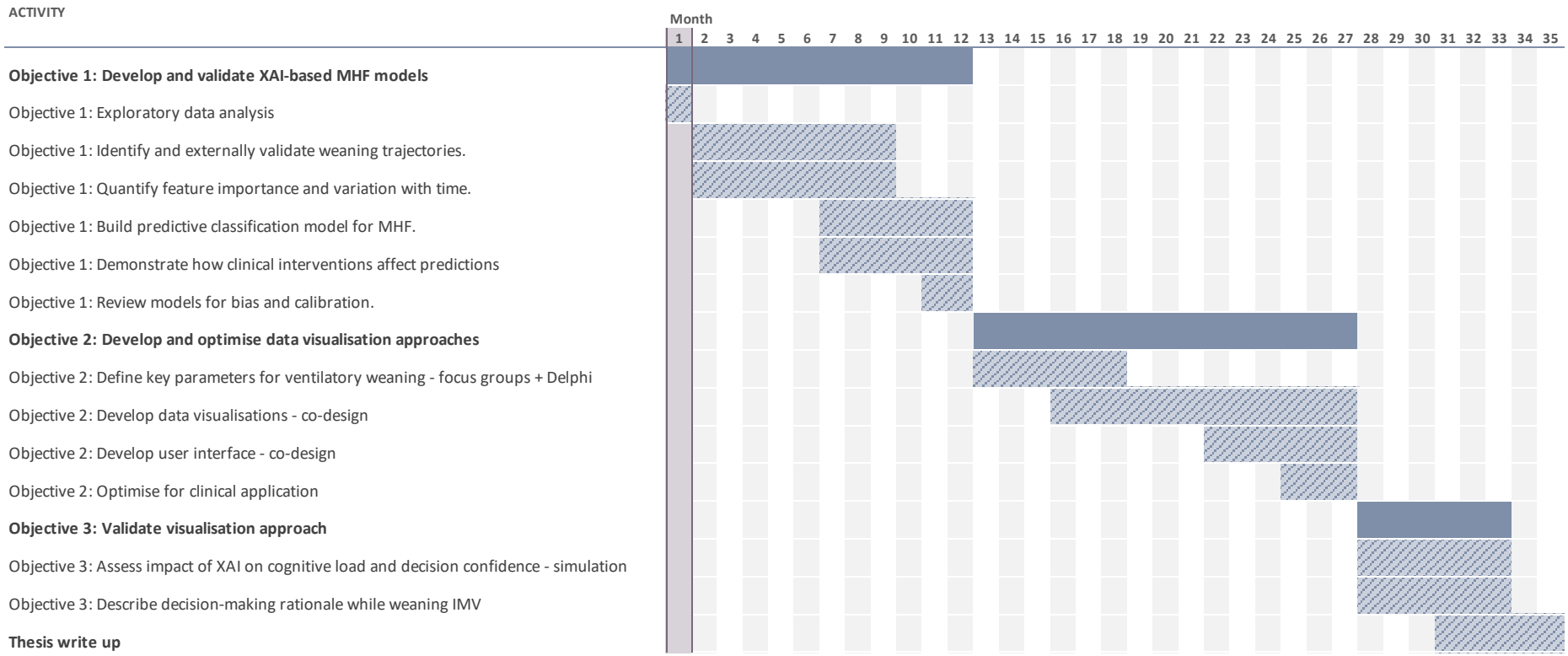
90. Hart SG, Field M. NASA-TASK LOAD INDEX (NASA-TLX); 20 YEARS LATER.

Appendix #2: Timeline_MM.pdf

BREATHE: Better Respiratory wEAning using Transparent macHine learning and data visualisation for invasivE mechanical ventilation

Select a period to highlight at right. A legend describing the charting follows.

1 | Plan Duration | Actual Start | % Complete | Actual (beyond plan) | % Complete (beyond plan)



Appendix #3: Funder Export_340376_SoECAT_BREATHE.xlsx